1 Intertemporal Trade and the Current Account Balance

One fundamental way open and closed economies differ is that an open economy can borrow resources from the rest of the world or lend them abroad. With the aid of loans from foreigners, an economy with a temporary income shortfall can avoid a sharp contraction of consumption and investment. Similarly, a country with ample savings can lend and participate in productive investment projects overseas. Resource exchanges across time are called *intertemporal trade*.

Much of the macroeconomic action in an open economy is connected with its intertemporal trade, which is measured by the current account of the balance of payments. The purpose of this chapter is to illustrate the basic economic principles that govern intertemporal trade patterns: when are countries foreign borrowers, when do they lend abroad, what role do government policies play, and what are the welfare implications of international capital-market integration? In the process, we take a first look at the key factors behind aggregate consumption and investment behavior and at the determination of world interest rates. We assume throughout that only one good exists on each date, the better to focus attention on aggregate international resource flows without introducing considerations related to changing intratemporal prices. A large part of international economics is, of course, concerned with relative domestic and international prices. As several later chapters illustrate, however, the macroeconomic roles these prices play are understood most easily if one starts off by abstracting from the complications they create.

1.1 A Small Two-Period Endowment Economy

You probably are familiar with the standard two-period microeconomic model of saving, due to Irving Fisher (1930). We begin by adapting Fisher's model to the case of a small open economy that consumes a single good and lasts for two periods, labeled 1 and 2. Although the model may seem simple, it is a useful building block for the more realistic models developed later. Our main goal in this section is to describe how a country can gain from rearranging the timing of its consumption through international borrowing and lending.

1.1.1 The Consumer's Problem

An individual i maximizes lifetime utility, U_1^i, which depends on period consumption levels, denoted c^i:

$$U_1^i = u(c_1^i) + \beta u(c_2^i), \qquad 0 < \beta < 1. \tag{1}$$

In this equation β is a fixed preference parameter, called the subjective discount or time-preference factor, that measures the individual's impatience to consume.

As usual, we assume that the period utility function $u(c^i)$ is strictly increasing in consumption and strictly concave: $u'(c^i) > 0$ and $u''(c^i) < 0$.[1]

Let y^i denote the individual's output and r the real interest rate for borrowing or lending in the world capital market on date 1. Then consumption must be chosen subject to the lifetime budget constraint

$$c_1^i + \frac{c_2^i}{1+r} = y_1^i + \frac{y_2^i}{1+r}. \tag{2}$$

This constraint restricts the present value of consumption spending to equal the present value of output. Output is perishable and thus cannot be stored for later consumption.[2]

We assume, as we shall until we introduce uncertainty about future income in Chapter 2, that the consumer bases decisions on *perfect foresight* of the future. This is an extreme assumption, but a natural one to make whenever the complexities introduced by uncertainty are of secondary relevance to the problem being studied. Perfect foresight ensures that a model's predictions are driven by its intrinsic logic rather than by ad hoc and arbitrary assumptions about how people form expectations. Unless the focus is on the economic effects of a particular expectational assumption per se, the deterministic models of this book therefore assume perfect foresight.[3]

To solve the problem of maximizing eq. (1) subject to eq. (2), use the latter to substitute for c_2^i in the former, so that the individual's optimization problem reduces to

$$\max_{c_1^i} u(c_1^i) + \beta u[(1+r)(y_1^i - c_1^i) + y_2^i].$$

The first-order condition for this problem is

1. Until further notice, we also assume that

$$\lim_{c^i \to 0} u'(c^i) = \infty.$$

The purpose of this assumption is to ensure that individuals always desire at least a little consumption in every period, so that we don't have to add formal constraints of the form $c^i \geq 0$ to the utility maximization problems considered later.

Whenever we refer to the subjective time-preference *rate* in this book, we will mean the parameter δ such that $\beta = 1/(1+\delta)$, that is, $\delta = (1-\beta)/\beta$.

2. At a positive rate of interest r, nobody would want to store output in any case. In section 1.2 we will see how this intertemporal allocation problem changes when output can be invested, that is, embodied in capital to be used in producing future output.

3. Even under the perfect foresight assumption we may sometimes loosely refer to an individual's "expectation" or (worse) "expected value" of a variable. You should understand that in a nonstochastic environment, these expectations are held with subjective certainty. Only when there is real uncertainty, as in later chapters, are expected values averages over nondegenerate probability distributions.

$$u'(c_1^i) = (1+r)\beta u'(c_2^i), \tag{3}$$

which is called an *intertemporal Euler equation*.[4] This Euler equation, which will recur in many guises, has a simple interpretation: at a utility maximum, the consumer cannot gain from feasible shifts of consumption between periods. A one-unit reduction in first-period consumption, for example, lowers U_1 by $u'(c_1^i)$. The consumption unit thus saved can be converted (by lending it) into $1+r$ units of second-period consumption that raise U_1 by $(1+r)\beta u'(c_2^i)$. The Euler equation (3) thus states that at an optimum these two quantities are equal.

An alternative and important interpretation of eq. (3) that translates it into language more closely resembling that of static price theory is suggested by writing it as

$$\frac{\beta u'(c_2^i)}{u'(c_1^i)} = \frac{1}{1+r}. \tag{4}$$

The left-hand side is the consumer's marginal rate of substitution of present (date 1) for future (date 2) consumption, while the right-hand side is the price of future consumption in terms of present consumption.

As usual, individual i's optimal consumption plan is found by combining the first-order condition (3) [or (4)] with the intertemporal budget constraint (2). An important special case is the one in which $\beta = 1/(1+r)$, so that the subjective discount factor equals the market discount factor. In this case the Euler equation becomes $u'(c_1^i) = u'(c_2^i)$, which implies that the consumer desires a flat lifetime consumption path, $c_1^i = c_2^i$. Budget constraint (2) then implies that consumption in both periods is $\bar{c}^i$, where

$$\bar{c}^i = \frac{[(1+r)y_1^i + y_2^i]}{2+r}. \tag{5}$$

1.1.2 Equilibrium of the Small Open Economy

We assume that all individuals in the economy are identical and that population size is 1. This assumption allows us to drop the individual superscript i and to identify per capita quantity variables with national aggregate quantities, which we denote by uppercase, nonsuperscripted letters. Thus, if C stands for aggregate consumption and Y for aggregate output, the assumption of a homogeneous population of size 1 implies that $c^i = C$ and $y^i = Y$ for all individuals i. Our assumed demographics simplify the notation by making the representative individual's first-order conditions describe aggregate dynamic behavior. The Euler equation (3), to take

4. The Swiss mathematician Leonhard Euler (1707–1783) served at one time as the court mathematician to Catherine the Great of Russia. The dynamic equation bearing his name arose originally in the problem of finding the so-called *brachistochrone*, which is the least-time path in a vertical plane for an object pulled by gravity between two specified points.

one instance, will also govern the motion of *aggregate* consumption under our convention.

We must keep in mind, however, that our notational shortcut, while innocuous in this chapter, is not appropriate in every setting. In later chapters we reintroduce individually superscripted lowercase quantity variables whenever consumer heterogeneity and the distinction between per capita and total quantities are important.

Since the only price in the model is the real interest rate r, and this is exogenously given to the small economy by the world capital market, national aggregate quantities are equilibrium quantities. That is, the small economy can carry out any intertemporal exchange of consumption it desires at the given world interest rate r, subject only to its budget constraint. For example, if the subjective and market discount factors are the same, eq. (5), written with C in place of c^i and Y in place of y^i, describes aggregate equilibrium consumption.

The idea of a representative national consumer, though a common device in modern macroeconomic modeling, may seem implausible. There are, however, three good reasons for taking the representative-consumer case as a starting point. First, several useful insights into the macroeconomy do not depend on a detailed consideration of household differences. An instance is the prediction that money-supply changes are neutral in the long run. Second, there are important cases where one can rigorously justify using the representative-agent model to describe aggregate behavior.[5] Finally, many models in international macroeconomics are interesting precisely because they assume differences between residents of different countries. Sometimes the simplest way to focus on these cross-country differences is to downplay differences within countries.

We have seen [in eq. (5)] that when $\beta = 1/(1+r)$, the time path of aggregate consumption is flat. This prediction of the model captures the idea that, other things the same, countries will wish to *smooth* their consumption. When the subjective time-preference rate and the market interest rate differ, the motivation to smooth consumption is modified by an incentive to *tilt* the consumption path. Suppose, for example, that $\beta > 1/(1+r)$ but $C_1 = C_2$. In this case the world capital market offers the country a rate of return that more than compensates it for the postponement of a little more consumption. According to the Euler equation (3), $u'(C_1)$ should exceed $u'(C_2)$ in equilibrium; that is, individuals in the economy maximize utility by arranging for consumption to rise between dates 1 and 2. The effects of a rise in

5. One does not need to assume literally that all individuals are identical to conclude that aggregate consumption will behave as if chosen by a single maximizing agent. Under well-defined but rather stringent preference assumptions, individual behavior can be aggregated exactly, as discussed by Deaton and Muellbauer (1980, ch. 6). We defer a formal discussion of aggregation until Chapter 5. For a perspective on ways in which the representative-agent paradigm can be misleading, however, see Kirman (1992).

r on initial consumption and on saving are rather intricate. We postpone discussing them until later in the chapter.

1.1.3 International Borrowing and Lending, the Current Account, and the Gains from Trade

Let's look first at how intertemporal trade allows the economy to allocate its consumption over time.

1.1.3.1 Defining the Current Account

Because international borrowing and lending are possible, there is no reason for an open economy's consumption to be closely tied to its current output. Provided all loans are repaid with interest, the economy's intertemporal budget constraint (2) is respected. In the special case $\beta = 1/(1+r)$, consumption is flat at the level $C_1 = C_2 = \bar{C}$ in eq. (5), but output need not be. If, for example, $Y_1 < Y_2$, the country borrows $\bar{C} - Y_1$ from foreigners on date 1, repaying $(1+r)(\bar{C} - Y_1)$ on date 2. Whenever date 2 consumption equals output on that date less the interest and principal on prior borrowing—that is, $C_2 = Y_2 - (1+r)(C_1 - Y_1)$—the economy's intertemporal budget constraint obviously holds true.

A country's *current account balance* over a period is the change in the value of its net claims on the rest of the world—the change in its net foreign assets. For example, in our initial simple model without capital accumulation, a country's first-period current account is simply national saving. (In section 1.2 we will see that in general a country's current account is national saving less domestic investment.) The current account balance is said to be in surplus if positive, so that the economy as a whole is lending, and in deficit if negative, so that the economy is borrowing.

Our definition of a country's current account balance as the increase in its net claims on foreigners may puzzle you if you are used to thinking of the current account as a country's *net exports* of goods and services (where "service" exports include the services of domestic capital operating abroad, as measured by interest and dividend payments on those assets). Remember, however, that a country with positive net exports must be acquiring foreign assets of equal value because it is selling more to foreigners than it is buying from them; and a country with negative net exports must be borrowing an equal amount to finance its deficit with foreigners. Balance-of-payments statistics record a country's net sales of assets to foreigners under its *capital account balance*. Because a payment is received from foreigners for any good or service a country exports, every positive item of its net exports is associated with an equal-value negative item in its capital account—namely, the associated payment from abroad, which is a foreign asset acquired. Thus, as a pure matter of accounting, the net export surplus and the capital account surplus sum identically to zero. Hence, the capital account surplus preceded by a

minus sign—the net *increase* in foreign asset holdings—equals the current account balance.

Despite this accounting equivalence, there is an important reason for focusing on the foreign asset accumulation view of the current account. It plainly shows that the current account represents trade *over* time, whereas the net exports view draws attention to factors determining gross exports and imports *within* a single time period. Those factors are far more than unimportant details, as we shall see in subsequent chapters, but to stress them at the outset would only obscure the basic principles of intertemporal trade.

To clarify the concept of the current account, let B_{t+1} be the value of the economy's net foreign assets at the end of a period t. The current account balance over period t is defined as $CA_t = B_{t+1} - B_t$. In general, the date t current account for a country with no capital accumulation or government spending is

$$CA_t = B_{t+1} - B_t = Y_t + r_t B_t - C_t, \tag{6}$$

where $r_t B_t$ is interest earned on foreign assets acquired previously. (This convention makes r_t the one-period interest rate that prevailed on date $t - 1$.)

1.1.3.2 Gross National Product and Gross Domestic Product

Equation (6) shows that a country's current account (or net export surplus) is the difference between its *total income* and its consumption. The national income of an economy is also called its *gross national product* (GNP) and is measured as the sum of two components: the value of the final output produced within its borders *and* net international factor payments. Here, these factor payments consist of interest and dividend earnings on the economy's net foreign assets, which are viewed as domestic capital operating abroad.[6] (In line with the definition of net exports given earlier, a country's earnings on its foreign assets are considered part of its national product despite the fact that this product is generated abroad.) In terms of our formal model, GNP over any period t is $Y_t + r_t B_t$, as just indicated.

The first component of national product, output produced within a country's geographical borders, is called *gross domestic product* (GDP). In the present model

6. Strictly speaking, national income equals national product plus net unrequited transfer payments from abroad (including items like reparations payments and workers' remittances to family members in other countries). Workers' remittances, which represent a payment for exported labor services, are not truly unrequited and are completely analogous to asset earnings, which are payments for capital services. We will treat them as such in section 1.5. In practice, however, national income accountants usually don't treat remittances as payments for service exports. The term "gross" in GNP reflects its failure to account for depreciation of capital—a factor absent from our theoretical model. When depreciation occurs, *net* national product (NNP) measures national income less depreciation. Empirical economists prefer to work with GNP rather than NNP data, especially in international comparisons, because actual national account estimates of depreciation are accounting measures heavily influenced by domestic tax laws. Reported depreciation figures therefore are quite unreliable and can differ widely from country to country. For the United States, a ballpark estimate of annual depreciation would be around 10 percent of GNP.

Table 1.1
GNP versus GDP for Selected Countries, 1990 (dollars per capita)

Country	GDP	GNP	Percent Difference
Australia	17,327	17,000	−1.9
Brazil	2,753	2,680	−2.7
Canada	21,515	20,470	−4.9
Saudi Arabia	5,429	7,050	29.9
Singapore	11,533	11,160	−3.2
United Arab Emirates	17,669	19,860	12.4
United States	21,569	21,790	1.0

Source: World Bank, *World Development Report 1992.*

GDP is Y_t. Typically the difference between national and domestic product is a rather small number, but for some countries, those which have amassed large stocks of foreign wealth or incurred substantial foreign debts, the difference can be significant. Table 1.1 shows several of these cases.

1.1.3.3 The Current Account and the Budget Constraint in the Two-Period Model

Our formulation of budget constraint (2) tacitly assumed that $B_1 = 0$, making $CA_1 = Y_1 - C_1$ on the formal model's date 1 (but not in general). By writing constraint (2) as a strict equality, we have also assumed that the economy ends period 2 holding no uncollected claims on foreigners. (That is, $B_3 = 0$. Obviously foreigners do not wish to expire holding uncollected claims on the home country either!) Thus,

$$\begin{aligned} CA_2 &= Y_2 + rB_2 - C_2 = Y_2 + r(Y_1 - C_1) - C_2 \\ &= -(Y_1 - C_1) = -B_2 = -CA_1, \end{aligned}$$

where the third equality in this chain follows from the economy's intertemporal budget constraint, eq. (2). Over any stretch of time, as over a single period, a country's cumulative current account balance is the change in its net foreign assets, but in our two-period model with zero initial and terminal assets, $CA_1 + CA_2 = B_3 - B_1 = 0$.

Figure 1.1 combines the representative individual's indifference curves with the intertemporal budget constraint (2), graphed as

$$C_2 = Y_2 - (1 + r)(C_1 - Y_1).$$

It provides a diagrammatic derivation of the small economy's equilibrium and the implied trajectory of its current account. (The figure makes no special assumption about the relation between β and $1 + r$.) The economy's optimal consumption choice is at point **C**, where the budget constraint is tangent to the highest

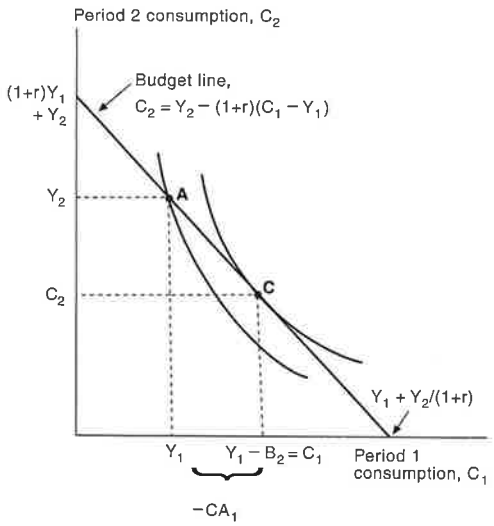


Figure 1.1
Consumption over time and the current account

attainable indifference curve. The first-period current account balance (a deficit in Figure 1.1) is simply the horizontal distance between the date 1 output and consumption points. As an exercise, the reader should show how to read from the figure's vertical axis the second-period current-account balance.

Economic policymakers often express concern about national current account deficits or surpluses. Our simple model makes the very important point that an unbalanced current account is not necessarily a bad thing. In Figure 1.1, for example, the country clearly does better running an unbalanced current account in both periods than it would if forced to set $C_1 = Y_1$ and $C_2 = Y_2$ (the autarky point **A**). Intertemporal trade makes possible a less jagged time profile of consumption. The utility gain between points **A** and **C** illustrates the general and classic insight that countries gain from trade.

Application: Consumption Smoothing in the Second Millennium B.C.

An early anecdote concerning the consumption-smoothing behavior underlying this chapter's model comes from the story of Joseph in the Book of Genesis. Scholars of the biblical period place the episode somewhere around 1800 B.C.

The Pharaoh of Egypt summoned Joseph, then an imprisoned slave, to interpret two dreams. In the first, seven plump cattle were followed and devoured by seven lean, starving cattle. In the second, seven full ears of corn were eaten by seven thin ears. After hearing these dreams, Joseph prophesied that Egypt would enjoy seven years of prosperity, followed by seven of famine. He recommended a consumption-smoothing strategy to provide for the years of famine, under which Pharaoh would appropriate and store a fifth of the grain produced during the years of plenty (Genesis 41:33–36). According to the Bible, Pharaoh embraced this plan, made Joseph his prime minister, and thereby enabled Joseph to save Egypt from starvation.

Why did Joseph recommend storing the grain (a form of domestic investment yielding a rate of return of zero before depreciation) rather than lending it abroad at a positive rate of interest? Cuneiform records of the period place the interest rate on loans of grain in Babylonia in a range of 20 to 33 percent per year and show clear evidence of international credit transactions within Asia Minor (Heichelheim 1958, pp. 134–135). At such high interest rates Egypt could have earned a handsome return on its savings. It seems likely, however, that, under the military and political conditions of the second millennium B.C., Egypt would have found it difficult to compel foreign countries to repay a large loan, particularly during a domestic famine. Thus storing the grain at home was a much safer course. The model in this chapter assumes, of course, that international loan contracts are always respected, but we have not yet examined mechanisms that ensure compliance with their terms. We will study the question in Chapter 6. ■

1.1.4 Autarky Interest Rates and the Intertemporal Trade Pattern

Diagrams like Figure 1.1 can illuminate the main factors causing some countries to run initial current account deficits while others run surpluses. The key concept we need for this analysis is the *autarky real interest rate*, that is, the interest rate that would prevail in an economy barred from international borrowing and lending.

Were the economy restricted to consume at the autarky point **A** in Figure 1.1, the only real interest rate consistent with the Euler eq. (3) would be the autarky interest rate r^A, defined by eq. (4) with outputs replacing consumptions:

$$\frac{\beta u'(Y_2)}{u'(Y_1)} = \frac{1}{1+r^A}. \tag{7}$$

This equation also gives the autarky price of future consumption in terms of present consumption.

Figure 1.1 shows that when the latter autarky price is below the *world* relative price of future consumption—which is equivalent to r^A being above r—future consumption is relatively cheap in the home economy and present consumption relatively expensive. Thus the home economy will "import" present consumption from abroad in the first period (by running a current account deficit) and "export" future consumption later (by repaying its foreign debt). This result is in accord

with the *principle of comparative advantage* from international trade theory, which states that countries tend to import those commodities whose autarky prices are high compared with world prices and export those whose autarky prices are comparatively low.[7] It is the opportunity to exploit these pretrade international price differentials that explains the gains from trade shown in Figure 1.1.

A rise in present output or a fall in future output lowers the autarky real interest rate: either event would raise desired saving at the previous autarky interest rate, but since the residents of a closed endowment economy cannot save more in the aggregate without lending abroad, r^A must fall until people are content to consume their new endowment. Similarly, greater patience (a rise in β) lowers r^A. By modifying Figure 1.1, you can check that when r^A is below the world interest rate r, the country runs a first-period current account surplus followed by a deficit, but still gains from trade.

It may come as a surprise that the existence of gains from intertemporal trade does not depend on the sign of the country's initial current account balance. The reason is simple, however. What produces gains is the chance to trade with someone different from oneself. Indeed, the greater is the difference, the greater the gain. The only case of no gain is the one in which, coincidentally, it happens that $r^A = r$.

This reasoning also explains how changes in world interest rates affect a country's welfare. In Figure 1.1 the economy reaps trade gains by borrowing initially because its autarky interest rate is above the world rate, r. Notice, however, that, were the world interest rate even lower, the economy's welfare after trade would be higher than in Figure 1.1. The basic reason for this welfare gain is that a fall in the world interest rate accentuates the difference between the home country and the rest of the world, increasing the gains from trade. A small rise in the world interest rate (one that doesn't reverse the intertemporal trade pattern) therefore harms a first-period borrower but benefits a first-period lender.

1.1.5 Temporary versus Permanent Output Changes

A suggestive interpretation of the preceding ideas leads to a succinct description of how alternative paths for output affect the current account.

The natural benchmark for considering the effects of changing output is the case $\beta = 1/(1+r)$. The reason is that, in this case, eq. (7) becomes

$$\frac{u'(Y_2)}{u'(Y_1)} = \frac{1+r}{1+r^A},$$

which implies that the *sole* factor responsible for any difference between the world and autarky interest rates is a changing output level.

7. For a detailed discussion, see Dixit and Norman (1980).

Imagine an economy that initially expects its output to be constant over time. The economy will plan on a balanced current account. But suppose Y_1 rises. If Y_2 does not change, the economy's autarky interest rate will fall below the world interest rate: a date 1 current account surplus will result as people smooth their consumption by lending some of their temporarily high output to foreigners. If Y_2 rises by the same amount as Y_1, however, the autarky interest rate does not change, and there is no current account imbalance. Alternatively, consumption automatically remains constant through time if people simply consume their higher output in both periods.

One way to interpret these results is as follows: *permanent* changes in output do not affect the current account when $\beta = 1/(1+r)$, whereas *temporary* changes do, temporary increases causing surpluses and temporary declines producing deficits. Likewise, a change in future expected output affects the sign of the current account in the same qualitative manner as an opposite movement in current output. We will generalize this reasoning to a many-period setting in the next chapter.

1.1.6 Adding Government Consumption

So far we have not discussed the role of a government. Government consumption is, however, easy to introduce.

Suppose government consumption per capita, G, enters the utility function additively, giving period utility the form $u(C) + v(G)$. This case is, admittedly, a simple one, but it suffices for the issues on which we focus. For now, it is easiest to suppose that the government simply appropriates G_t in taxes from the private sector for $t = 1, 2$. This policy implies a balanced government budget each period (we will look at government deficits in Chapter 3). The representative private individual's lifetime budget constraint is thus

$$C_1 + \frac{C_2}{1+r} = Y_1 - G_1 + \frac{Y_2 - G_2}{1+r}. \tag{8}$$

Government spending also enters the date t current account identity, which is now

$$CA_t = B_{t+1} - B_t = Y_t + r_t B_t - C_t - G_t.$$

The new feature here is that both government and private consumption are subtracted from national income to compute the current account. (Plainly we must account for *all* domestic expenditure—public as well as private—to reckon how much a country as a whole is saving.)

Since G is beyond the private sector's control we can follow the same steps as in section 1.1.1 to conclude that the Euler equation (3) remains valid. Indeed, introducing government consumption as we have done here is equivalent

to relabeling the private sector's endowment Y as output net of government consumption, $Y - G$.

How do government consumption decisions affect the current account? A natural benchmark once again is the case in which $\beta = 1/(1+r)$, and output is constant at $Y_1 = Y_2 = \bar{Y}$. Absent government consumption, private consumption would be constant in this case at $\bar{C} = \bar{Y}$, with the current account balanced. Suppose, however, that $G_1 > 0$ while $G_2 = 0$. Now the private sector will want to borrow against its relatively high second-period after-tax income to shift part of the burden of the temporary taxes to the future. The country therefore will run a deficit in period 1 and a surplus in period 2.

Replacing Y with $Y - G$ in eq. (5) implies that in the preceding example

$$\bar{C} = \frac{[(1+r)(\bar{Y} - G_1) + \bar{Y}]}{2+r} = \bar{Y} - \frac{(1+r)G_1}{2+r}.$$

Government consumption in period 1 lowers private consumption, but by an amount *smaller* than G_1. The reason is that the government consumption is temporary: it drops to zero in period 2. Thus the current account equation presented earlier in this subsection (recall that $B_1 = 0$ here) implies that

$$CA_1 = \bar{Y} - \bar{C} - G_1 = -\frac{G_1}{2+r} < 0.$$

In contrast, suppose that $G_1 = G_2 = \bar{G}$. Then consumption is constant at $\bar{C} = \bar{Y} - \bar{G}$ in both periods, and the current account is balanced always. Government consumption affects the current account here only to the extent that it tilts the path of private *net* income.

* 1.1.7 A Digression on Intertemporal Preferences

Equation (1) assumes the representative individual's preferences are captured by a very particular lifetime utility function rather than an unrestricted function $U_1 = U(C_1, C_2)$. In eq. (1) consumption levels for different dates enter additively; moreover, the *period* utility function $u(C)$ is constant over time. With consumption occurring over T rather than just two periods, the natural generalization of utility function (1) is

$$U_1 = \sum_{t=1}^{T} \beta^{t-1} u(C_t). \tag{9}$$

Preferences that can be represented by an additive lifetime utility function are called *intertemporally additive preferences*. The key property implied by intertemporal additivity is that the marginal rate of substitution between consumption on any two dates t and s [equal to $\beta^{s-t}u'(C_s)/u'(C_t)$ for the preferences described by eq. (9)] is independent of consumption on any third date. This property is restrictive (provided $T > 2$, of course). It rules out certain kinds of intertemporal consumption dependencies, such as complementarity between total consumption levels in different periods. Such dependencies are at the heart of recent models of habit persistence in aggregate consumption.[8]

Although we will discuss particular alternative assumptions on tastes at several points in the book, the assumption of intertemporally additive preferences with an unvarying period utility function will form the backbone of our formal analysis. There are several reasons for this choice:

1. It is true that some types of goods, such as refrigerators and automobiles, are *durable* goods typically consumed over many periods rather than just one. This type of consumption linkage, however, is fundamentally technological. By defining utility over the flow of services from durables, and by imputing their rental cost, one can easily incorporate such goods within the umbrella of intertemporally additive preferences. (We show this in Chapter 2.)

2. For some types of goods, consumption at one point in time clearly does influence one's utility from consuming in closely neighboring periods. After eating a large meal, one is less inclined to want another an hour later. The time intervals of aggregation we look at in macroeconomic data, however, typically are measured in months, quarters, or years, periods over which many types of intertemporal dependencies fade.

3. Admittedly, even over long periods, habit persistence can be important. Drug addiction is an extreme example; watching television is a closely related one. In macroeconomics, however, one should think of preferences as being defined over consumption variables that really represent aggregate spending on a wide array of different goods. While we may have some intuition about the persistence effects of consuming certain items, it is harder to see obvious and quantitatively significant channels through which the *totality* of consumption has long-lived persistence effects.

4. One can think of some types of goods that most individuals would prefer to consume only once, such as marriage services. But even though consumption of such services is lumpy for an individual, it is relatively smooth in the aggregate.

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8. If $G(\cdot)$ is any monotonically increasing function, then the utility function

$$G\left[\sum_{t=1}^{T} \beta^{t-1} u(C_t)\right]$$

naturally represents the same preferences as U_1 does, i.e., a monotonically increasing transformation of the lifetime utility function does not affect the consumer's underlying preference ordering over different consumption paths. Intertemporally additive preferences take the general form $G[u_1(C_1) + \cdots + u_T(C_T)]$ (with period utility functions possibly distinct). They also go by the name *strongly intertemporally separable* preferences. For further discussion of their implications, see Deaton and Muellbauer (1980, ch. 5.3).

People get married all the time. Similarly, people may take vacation trips only at infrequent intervals, but this is not the case in the aggregate. (Seasonality can be important in either of these examples, but such effects are easily dealt with.)

5. Fundamentally, a very general intertemporally nonadditive utility function would yield few concrete behavioral predictions. If consumptions on different dates are substitutes, one gets dramatically different results from the case in which they are complements. Because maximal generality would lead to an unfalsifiable macroeconomic theory with little empirical content, macroeconomists have found it more fruitful to begin with a tractable basic setup like eq. (9), which has very sharp predictions. The basic setup can then be amended in parsimonious and testable ways if its implications seem counterintuitive or counterfactual.

6. In any event, while empirical research has raised interesting questions about the simplest time-additive preference model, it does not yet clearly point to a superior nonadditive alternative.

1.2 The Role of Investment

Historically, one of the main reasons countries have borrowed abroad is to finance productive investments that would have been hard to finance out of domestic savings alone. In the nineteenth century, the railroad companies that helped open up the Americas drew on European capital to pay laborers and obtain rails, rolling stock, and other inputs. To take a more recent example, Norway borrowed extensively in world capital markets to develop its North Sea oil resources in the 1970s after world oil prices shot up.

So far we have focused on consumption smoothing in our study of the current account, identifying the current account with national saving. In general, however, the current account equals saving minus investment. And because, in reality, investment usually is much more volatile than saving, to ignore investment is to miss much of the action.

1.2.1 Adding Investment to the Model

Let's modify our earlier model economy to allow for investment. We now assume that output is produced using capital, which, in turn, can be accumulated through investment. The production function for new output in either period is

$$Y = F(K). \tag{10}$$

As usual production is strictly increasing in capital but subject to diminishing marginal productivity: $F'(K) > 0$ and $F''(K) < 0$. Furthermore, output cannot be produced without capital: $F(0) = 0$. We will think of the representative consumer as having the additional role of producer with direct access to this technology.[9]

A unit of capital is created from a unit of the consumption good. This process is reversible, so that a unit of capital, after having been used to produce output, can be "eaten." You may find these assumptions unrealistic, but they help us sidestep some technical issues that aren't really central here. One key simplification due to our assumptions is that the relative price of capital goods in terms of consumption always equals 1.

Introducing investment requires that we rethink the budget constraints individuals face, because now saving can flow into capital as well as foreign assets. Total domestic private wealth at the end of a period t is now $B_{t+1} + K_{t+1}$, the sum of net foreign assets B_{t+1} and the stock of domestic capital K_{t+1}.[10]

How is capital investment reflected in the date t current account? The stock of capital K_{t+1} accumulated through the end of period t is the sum of preexisting capital K_t and new investment during period t, I_t (we ignore depreciation of capital):

$$K_{t+1} = K_t + I_t. \tag{11}$$

Nothing restricts investment to be nonnegative, so eq. (11) allows people to eat part of their capital.

Next, the change in total domestic wealth, national saving, is

$$B_{t+1} + K_{t+1} - (B_t + K_t) = Y_t + r_t B_t - C_t - G_t.$$

Finally, rearranging terms in this equation and substituting (11) shows that the current account surplus is

$$CA_t = B_{t+1} - B_t = Y_t + r_t B_t - C_t - G_t - I_t. \tag{12}$$

A very useful way to interpret the preceding current account identity is to label national saving as S_t:

$$S_t \equiv Y_t + r_t B_t - C_t - G_t. \tag{13}$$

9. As we discuss in later chapters, it is reasonable to think of labor as being an additional production input alongside capital. A production function of the form (10) still is valid as long as labor is supplied inelastically by the individual producer. We assume

$$\lim_{K \to 0} F'(K) = \infty$$

to ensure a positive capital stock.

10. It is simplest to suppose that all domestic capital is owned by domestic residents. The statement that total domestic wealth equals $B + K$ is true even when foreigners own part of the domestic capital stock, however, because domestic capital owned by foreigners is subtracted in calculating *net* foreign assets B. As long as perfect foresight holds, so that the ex post returns to assets are equal, the ownership of the domestic capital stock is irrelevant. The ownership pattern is not irrelevant, as we see later, when unexpected shocks can occur.

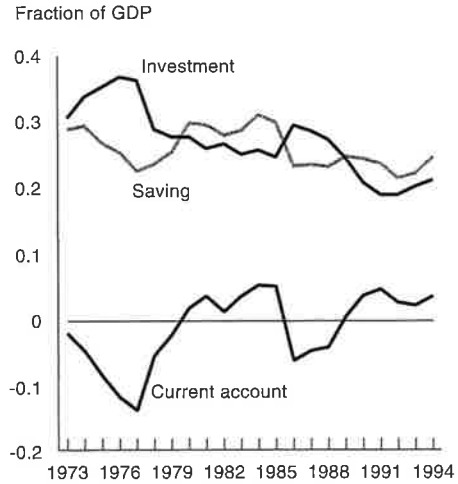


Figure 1.2
Norway's saving-investment balance, 1973–94. (*Source:* OECD)

Then eq. (12) states that in an economy with investment,

$$CA_t = S_t - I_t. \tag{14}$$

National saving in excess of domestic capital formation flows into net foreign asset accumulation.

The saving-investment identity (14) discloses that the current account is fundamentally an *intertemporal* phenomenon. Simple as it is, the identity $CA = S - I$ is vital for analyzing how economic policies and disturbances change the current account. Will a protective tariff, often imposed to improve the current account, succeed in its aim? The answer cannot be determined from partial-equilibrium reasoning, but ultimately depends instead on how the tariff affects saving and investment.

Figure 1.2 returns to the Norwegian case mentioned at the start of this section, graphing recent data on saving, investment, and the current account. In the mid-1970s, the Norwegian current account registered huge deficits, touching -14 percent of GDP in 1977. In an accounting sense, higher energy-sector investment is "responsible" for much of the deficit, although saving simultaneously fell in the mid-1970s, possibly in anticipation of higher future oil revenues. Subsequent surpluses through 1985, reflecting higher saving and lower investment, enabled Norway to repay much of the debt incurred in the 1970s.

The Norwegian data illustrate an important point. The saving-investment identity is a vital analytical tool, but because CA, S, and I are jointly determined endogenous variables that respond to common exogenous shocks, it may be misleading to identify a specific ex post investment or saving shift as the "cause" of a current account change. Our model with investment will show how various exogenous shocks can simultaneously affect all three variables in the saving-investment identity.

1.2.2 Budget Constraint and Individual Maximization

To derive the intertemporal budget constraint analogous to eq. (8) when there is both government spending and investment, we simply add the asset-accumulation identities for periods 1 and 2. For period 1, current account eq. (12) gives

$$B_2 = Y_1 - C_1 - G_1 - I_1$$

(recall that $B_1 = 0$). For period 2, eq. (12) gives

$$-B_2 = Y_2 + rB_2 - C_2 - G_2 - I_2$$

(recall that $B_3 = 0$). Solve this equation for B_2, and substitute the result into the equation that precedes it. One thereby arrives at the intertemporal budget constraint

$$C_1 + I_1 + \frac{C_2 + I_2}{1+r} = Y_1 - G_1 + \frac{Y_2 - G_2}{1+r}. \tag{15}$$

Now it is the present value of consumption *plus* investment that is limited by the present value of output.

In this economy with investment, a representative individual maximizes $U_1 = u(C_1) + \beta u(C_2)$ subject to eq. (15), where eq. (10) replaces Y with $F(K)$ and eq. (11) is used to replace I with the change in K. To simplify further, observe that people will never wish to carry capital past the terminal period 2. Thus capital K_2 accumulated in period 1 will be consumed at the end of period 2 and K_3 will be zero, implying that

$$I_2 = K_3 - K_2 = 0 - K_2 = -K_2.$$

Using eq. (15) to eliminate C_2 from U_1 therefore transforms the individual's problem to

$$\max_{C_1, I_1} u(C_1)$$
$$+ \beta u\left\{(1+r)\left[F(K_1) - C_1 - G_1 - I_1\right] + F(I_1 + K_1) - G_2 + I_1 + K_1\right\}. \tag{16}$$

(K_1 is given by history and is not subject to choice on date 1.) The two corresponding first-order conditions are the Euler equation (3) and

$$F'(K_2) = r, \tag{17}$$

where we have used the identity $K_2 = K_1 + I_1$.

An extra unit of output invested on date 1 can be fully consumed, together with its marginal contribution to output, $F'(K_2)$, on date 2. Equation (17) says that

Box 1.1
Nominal versus Real Current Accounts

Our use in Figure 1.2 of data from official national income and product accounts raises an important measurement problem that you should recognize as you read this book. Unfortunately, the problem is easier to understand than to cure, so in most cases we reluctantly continue to rely on the official data.

Ideally, the current account should measure the change in an economy's net *real* claims on foreigners. In practice, however, government statistical agencies measure the current account and GDP by adding up the values of transactions measured in *nominal* terms, that is, in units of domestic money. This practice poses no conceptual hazards when money has a stable value in terms of real output, but, for reasons we will understand better after learning about monetary economics in Chapters 8–10, real-world economies are almost always afflicted by at least some *price-level inflation*, a tendency for the money prices of all goods and services to rise over time. Such inflation would not be a problem if all international borrowing and lending involved the exchange of output-indexed bonds, as our theoretical model assumes. But most bonds traded between countries have returns and face values that are contracted in terms of currencies, implying that inflation can affect their real values.

A hypothetical example illustrates the problem. Suppose United States GDP is $7 trillion dollars and the U.S. net foreign debt is $700 billion. Suppose also that all international debts are linked to dollars, that the interest rate on dollar loans is 10 percent per year, and that U.S. GDP equals the sum of consumption, investment, and government spending. Under these assumptions the U.S. Department of Commerce would report the current account balance as the nominal interest outflow on net foreign assets, or $(0.1) \times (\$700 \text{ billion}) = \$70$ billion. So measured, the current account deficit is 1 percent of GDP.

Suppose, however, that all dollar prices are rising at 5 percent per year. Over the course of the year, the U.S. external debt declines in real value by $(0.05) \times (\$700 \text{ billion}) = \$35$ billion as a result of this inflation. Thus the dollar value of the change in U.S. *real* net foreign assets is not $70 billion, but $70 billion $-\$35$ billion $= \$35$ billion. This smaller number divided by GDP, equal to 0.5 percent, shows the change in the economy's real net foreign assets as a fraction of real output. Naive use of nominal official numbers makes the deficit look twice as large relative to GDP as it really is!

While it was easy to measure the current account correctly in our example, doing so in practice is much harder. International financial transactions are denominated in many currencies. Changes in currency exchange rates as well as in national money price levels therefore enter into the real current account, but, because the currency composition of a country's net foreign debt is difficult to monitor, accurate correction is problematic. Many internationally held assets, such as stocks, long-term bonds, and real estate, can fluctuate sharply in value. Accounting for these price changes involves similar problems.

Caveat emptor. Unless otherwise stated, the ratios of the current account to output that you encounter in this book are the rough approximations one gleans from official national accounts. The same is true of related wealth flows, such as saving-to-output ratios.

period 1 investment should continue to the point at which its marginal return is the same as that on a foreign loan. A critical feature of eq. (17) is its implication that the desired capital stock is *independent* of domestic consumption preferences! Other things equal, wouldn't a less patient country, one with a lower value of β, wish to invest less? Not necessarily, if it has access to perfect international capital markets. A country that can borrow abroad at the interest rate r never wishes to pass up investment opportunities that offer a net rate of return above r.

Several key assumptions underpin the separation of investment from consumption decisions in this economy. First, the economy is *small*. The saving decisions of its residents don't change the interest rate at which investment projects can be financed in the world capital market. Second, the economy produces and consumes a single tradable good. When the economy produces nontraded goods, as in some of Chapter 4's models, consumption shifts can affect investment. Third, capital markets are free of imperfections that might act to limit borrowing. We shall see later (in Chapter 6) that when factors such as default risk restrict access to international borrowing, national saving can influence domestic investment.[11]

In the present setup, investment is independent of government consumption as well. In particular, government consumption does not crowd out investment in a small open economy facing a perfect world capital market.

1.2.3 Production Possibilities and Equilibrium

Let's assume temporarily that government consumption is zero in both periods. Then Figure 1.3 shows how the current account is determined when there is investment. To the information in Figure 1.1, Figure 1.3 adds an *intertemporal production possibilities frontier* (PPF) showing the technological possibilities available in autarky for transforming period 1 consumption into period 2 consumption. The PPF is described by the equation

$$C_2 = F\left[K_1 + F(K_1) - C_1\right] + K_1 + F(K_1) - C_1. \tag{18}$$

What does this equation imply about the PPF's position and shape? If the economy chose the lowest possible investment level on date 1 by eating all its inherited capital immediately (setting $I_1 = -K_1$), it would enjoy the highest date 1 consumption available in autarky, $C_1 = K_1 + F(K_1)$. In this case date 2 consumption

11. Once we allow for uncertainty, as in Chapters 2 and 5, restrictions on the tradability of certain assets also can upset the separation of investment from consumption. We have not yet introduced labor as an explicit factor of production, but if the supply of labor influences the marginal product of capital, the separation can also fail when consumption and labor effort enter the period utility function in a nonadditive manner.

Even if consumption shifts don't alter investment, the converse proposition need *not* be true! As budget constraint (15) shows, investment enters the consumer's budget constraint in equilibrium, so in general factors that shift domestic investment can affect national saving too.

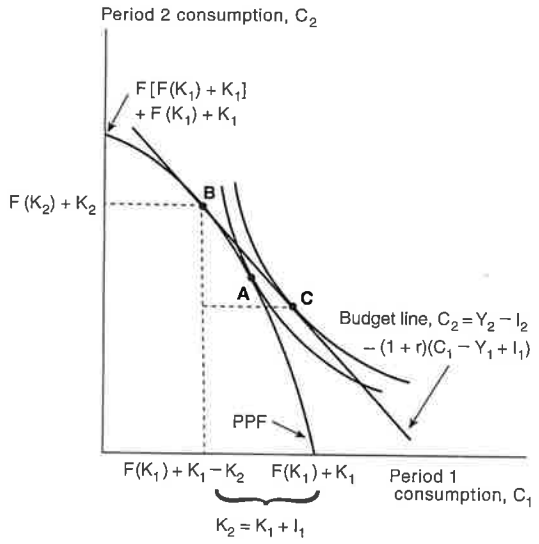


Figure 1.3
Investment and the current account

would be at its lowest possible level, $C_2 = F(0) + 0 = 0$. The resulting point is the PPF's horizontal intercept in Figure 1.3. At the other extreme, the economy could invest all output on date 1 and eat no inherited capital K_1. This decision would set $C_1 = 0$, $I_1 = F(K_1)$, $K_2 = K_1 + F(K_1)$, and $C_2 = F\left[K_1 + F(K_1)\right] + K_1 + F(K_1)$, the last being the highest date 2 consumption available in autarky. The PPF's vertical intercept in Figure 1.3 shows the allocation just described. In between, the PPF's slope is obtained from (18) by differentiation:

$$\frac{dC_2}{dC_1} = -[1 + F'(K_2)].$$

Capital's diminishing marginal productivity makes the PPF strictly concave, as shown.[12]

Point **A** in Figure 1.3 is the autarky equilibrium. There, the PPF is tangent to the highest indifference curve the economy can reach without trade. The common slope of the two curves at **A** is $-(1 + r^A)$, where r^A, as earlier, is the autarky real interest rate. All three closed-economy equilibrium conditions hold at point **A**. First, producer maximization: investment decisions are efficient, given r^A [that

12. To test your understanding, show that the second derivative $d^2C_2/dC_1^2 = F''(K_2) < 0$.

is, condition (17) holds for r^A]. Second, consumer maximization: the intertemporal Euler condition holds, again given r^A. Third, output-market clearing: consumption and investment sum to output in both periods. You can see that markets clear at **A** by observing that in autarky K_2 equals the distance between C_1 and $Y_1 + K_1$ along the horizontal axis, so that $C_1 + I_1 = Y_1$, whereas eq. (18) can be written as $C_2 = Y_2 + K_2 = Y_2 - I_2$.

In Figure 1.3, the economy faces a *world* interest rate, r, lower than the autarky rate r^A implied by the dual tangency at point **A**. Thus, at **A**, the marginal domestic investment project offers a rate of return above the world cost of borrowing. The opportunity to trade across periods with foreigners lets domestic residents gain by investing more and producing at point **B**, through which the economy's new budget line passes. Production at **B** maximizes the present value of domestic output (net of investment) by placing the economy on the highest feasible budget line at world prices. Consuming at point **C** gives the economy the highest utility level it can afford.

The horizontal distance between points **A** and **B** is the extra investment generated by opening the economy to the world capital market. The horizontal distance between points **A** and **C** shows the extra first-period consumption that trade simultaneously allows. Since total first-period resources $Y_1 + K_1$ haven't changed, the sum of these two horizontal distances—the distance from **B** to **C**—is the first-period current account deficit.

The utility curve through point **C** lies above the one through point **A**. The distance between them measures the gains from trade. In Figure 1.1, trade gains were entirely due to smoothing the time path of consumption. In Figure 1.3 there is an additional source of gain, the change in the economy's production point from **A** to **B**.

Had the world interest rate r been *above* r^A rather than below it, the country would have run a first-period current account surplus but still enjoyed gains from intertemporal trade, as in the pure endowment case studied earlier.

1.2.4 The Model with Government Consumption

In section 1.2.3 we assumed away government consumption. Now we reinstate it in our graphical analysis.

A glance at eqs. (15) (for the individual's intertemporal budget constraint) and (18) (for the PPF) shows how changes in government consumption affect the graphs of these two relations between C_2 and C_1: both are shifted vertically downward by the amount of the increase in G_2 and horizontally leftward by the amount of the increase in G_1.

In understanding the difference that government consumption can make, it helps intuition to begin with an economy in which government consumption is always zero and the current account is in balance, so that consumption and production are

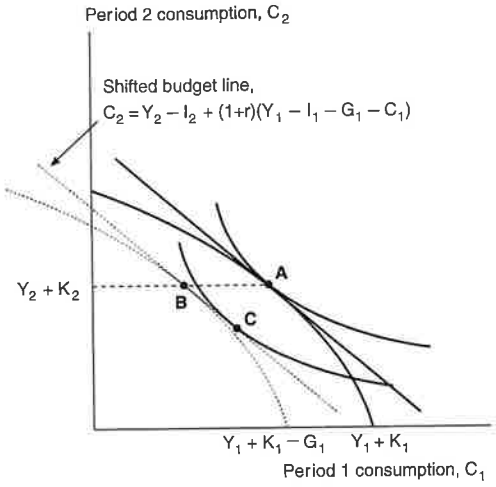


Figure 1.4
Government consumption and the current account

at point **A** in Figure 1.4. Compare this economy with an otherwise identical one in which $G_1 > 0$ while G_2 remains at zero. In the second economy, both the PPF and the budget constraint have shifted to the left by an amount G_1, and the economy's production point is **B**, implying the same investment level as at **A**.[13] Notice, however, that as long as consumption is a *normal* good on both dates, consumers will not wish to consume at **B**, for this choice would imply an unchanged C_2. Instead, they respond to a lower lifetime income level by reducing consumption on *both* dates and choosing consumption point **C**, which is southeast of **B** on the new budget constraint.

We conclude from Figure 1.4 that, other things equal, an economy with disproportionately high period 1 government consumption will have a current account deficit in that period. When government consumption is expected to occur on the future date 2 instead, the current account will be in surplus on date 1. Both predictions are explained by individuals' efforts to spread the burden of higher taxes over both periods of life through borrowing or lending abroad.

13. It may seem odd at first glance that in autarky a rise in G_1 alone reduces the maximal output available for private consumption on date 2 as well as on date 1. Recall, however, that when private consumption is zero on date 1, investment is lower by G_1 in autarky, so the maximal date 2 consumption available to the autarkic economy is only $F(K_1 + Y_1 - G_1) + K_1 + Y_1 - G_1$.

1.3 A Two-Region World Economy

Until now we have focused on a country too small to affect the world interest rate. In this section we show how the world interest rate is determined and how economic events in large regions are transmitted abroad.

1.3.1 A Global Endowment Economy

Let us start by abstracting from investment again and assuming a world of two regions or countries, called Home and Foreign, that receive exogenously determined endowments on dates 1 and 2. The two economies have parallel structures, but symbols pertaining to Foreign alone are marked by asterisks. We also omit government spending, which operates precisely as a reduction in output in our model.

Equilibrium in the global output market requires equal supply and demand on each date $t = 1, 2$,

$$Y_t + Y_t^* = C_t + C_t^*.$$

Equivalently, subtracting world consumption from both sides in this equation implies world saving must be zero for $t = 1, 2$,

$$S_t + S_t^* = 0.$$

Since there is so far no investment in the model, this equilibrium condition is the same as $CA_t + CA_t^* = 0$. We can simplify further by recalling that when there are only two markets, output today and output in the future, we need only check that one of them clears to verify general equilibrium (Walras's law). Thus the world economy is in equilibrium if

$$S_1 + S_1^* = 0. \tag{19}$$

Figure 1.5 shows how the equilibrium world interest rate is determined for given present and future endowments. In this case a country's date 1 saving depends only on the interest rate it faces. Curve **SS** shows how Home saving depends on r and curve **S***S* does the same for Foreign. We will probe more deeply into the shapes of the saving schedules in a moment, but for now we ask you to accept them as drawn in Figure 1.5.

In Figure 1.5, the equilibrium world interest rate makes Home's lending, measured by the length of segment **AB**, equal to Foreign's borrowing, measured by the length of **B***A*. The equilibrium world interest rate r must lie between the two autarky rates:

$$r^A < r < r^{A*}.$$

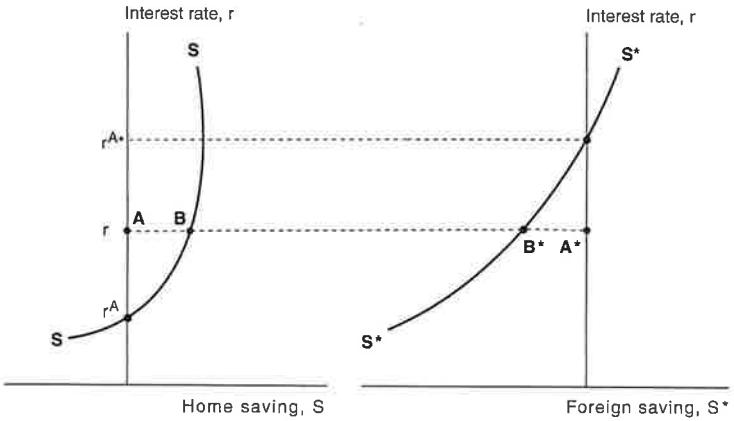


Figure 1.5
Global exchange equilibrium

In Home, the rise in the interest rate from its autarky level encourages saving, leading to positive Home saving of **AB**. Home's first-period current account surplus also equals **AB**. Foreign's situation is the reverse of Home's, and it runs a current account deficit of **B*A***. (Because Home and Foreign face the same world interest rate, we do not mark it with an asterisk.) The intertemporal trade pattern naturally conforms to the comparative advantage principle.

It is easy to see in Figure 1.5 how changes in exogenous variables alter the world interest rate and international capital flows. A ceteris paribus increase in Home's date 1 output, as we know, leads the country to raise its saving at a given rate of interest. As a result, **SS** shifts to the right. Plainly, the new equilibrium calls for a lower world interest rate, higher Home lending on date 1, and higher Foreign borrowing. Other things equal, higher date 2 output in Home shifts **SS** leftward, with opposite effects. Changes in Foreign's intertemporal endowment pattern work similarly, but through a shift of the Foreign saving schedule **S*S***.

An important normative issue concerns the international distribution of the benefits of economic growth. Is a country helped or hurt by an increase in trading partners' growth rates? To be concrete, suppose Home's date 2 output Y_2 rises, so that **SS** shifts leftward and the world interest rate rises. Because Foreign finds that the terms on which it must borrow have worsened, Foreign is actually worse off. Home, conversely, benefits from a higher interest rate for the same reason: the terms on which it lends to Foreign have improved. Thus, alongside the primary gain due to future higher output, Home enjoys a secondary gain due to the induced change in the intertemporal relative price it faces.

An interest-rate increase improves the *intertemporal terms of trade* of Home, which is "exporting" present consumption (through a date 1 surplus), while worsening those of Foreign, which is "importing" present consumption (through a date 1 deficit). In general, a country's terms of trade are defined as the price of its exports in terms of its imports. Here, $1 + r$ is the price of present consumption in terms of future consumption, that is, the price of a date 1 surplus country's export good in terms of its import good. As in static trade theory, a country derives a positive welfare benefit when its terms of trade improve and a negative one when they worsen.

It may seem reasonable to suppose that if Home's date 1 output Y_1 rises Home must benefit. In this case, however, the last paragraph's reasoning works in reverse. Because the world interest rate falls, Home's terms of trade worsen and counteract the primary benefit to Home of higher date 1 output. (Part of Home's benefit is exported abroad, and Foreign's welfare unambiguously rises.) Indeed, the terms-of-trade effect can be so big that higher date 1 output for Home actually worsens its lot. This paradoxical outcome has been dubbed *immiserizing growth* by Bhagwati (1958).

Application: War and the Current Account

Nothing in human experience is more terrible than the misery and destruction caused by wars. Their high costs notwithstanding, wars do offer a benefit for empirical economists. Because wars have drastic consequences for the economies involved, usually are known in advance to be temporary, and, arguably, are exogenous, they provide excellent "natural experiments" for testing economic theories.

Wartime data can have drawbacks as well. During wars, market modes of allocation may be supplemented or replaced by central economic planning. Because price controls and rationing are common, data on prices and quantities become hard to interpret in terms of market models. Matters are even worse when it comes to testing open-economy models, as wars inevitably bring tighter government control over capital movements and trade. Sometimes the normal data collecting and processing activities of statistical agencies are disrupted.

One way to reduce some of these problems is to focus on data from before the 1930s, when governments decisively turned away from laissez-faire in attempts to shield their economies from the worldwide Great Depression. Although pre-1930s data can be of uneven quality compared with modern-day numbers, they have been surprisingly useful in evaluating modern theories. We illustrate the use of historical data by looking briefly at the effects on both bystanders and participants of some early twentieth-century wars.

Sweden did not directly participate in World War I, while Japan took part only peripherally. Current-account data for the war's 1914–18 span are available

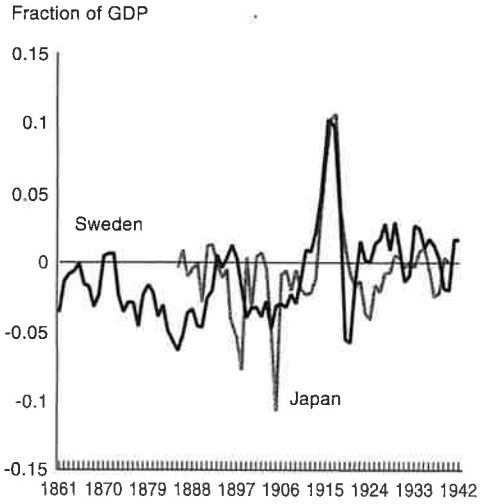


Figure 1.6
Current accounts of Japan and Sweden, 1861–1942

for both countries. What does our model predict about the effect of a foreign war on nonbelligerents? Return to Figure 1.5, interpreting "Home" as the warring portion of the world. For inhabitants of Home the war represents a situation in which the output available for private consumption has exogenously become much lower in the present than in the future. In response, Home lowers its saving at every interest rate, causing **SS** to shift to the left. Home's current account surplus falls (and may become a deficit), and the world interest rate rises. In Foreign, the region still at peace, the rise in the world interest rate causes a rise in saving and an improved current account balance (perhaps even a surplus).

Figure 1.6, which graphs current account data for Japan and Sweden, is consistent with the prediction that nonparticipants should run surpluses during large foreign wars. In both countries there is an abrupt shift from secular deficit toward a massive surplus reaching 10 percent of national product. The huge surpluses disappear once the war is over.

What is the evidence that belligerents do wish to borrow abroad? Foreign financing of wars has a long history; over the centuries, it has helped shape the institutions and instruments of international finance. From the late seventeenth century through the end of the Napoleonic Wars, lenders in several other continental countries underwrote Britain's military operations abroad. As far back as the first half of the fourteenth century, Edward III of England invaded France with the help of

Table 1.2
Japan's Gross Saving and Investment During the Russo-Japanese War (fraction of GDP)

Year	Saving/GDP	Investment/GDP
1903	0.131	0.136
1904	0.074	0.120
1905	0.058	0.168
1906	0.153	0.164

loans from Italian bankers. Edward's poor results in France and subsequent refusal to honor his foreign debts illustrate a potential problem for tests of the hypothesis that wars worsen the current account. Even though a country at war may wish to borrow, why should lenders respond when a country's ability to repay may be impaired even in the event of victory? The prospect that borrowers default can limit international capital flows, as we discuss in greater detail in Chapter 6. But intergovernment credits often are extended in wartime, and private lenders may stay in the game, too, if the interest rates offered them are high enough to compensate for the risk of loss.

Japan's 1904–1905 conflict with Russia offers a classic example of large-scale borrowing to finance a war. In February 1904, tensions over Russia's military presence in Manchuria and its growing influence in Korea erupted into open hostilities. Public opinion on the whole favored Japan, but Russia's superiority in manpower and other resources led more sober commentators to predict that the great power would beat its upstart challenger in the long run. These predictions quickly faded as Japan's naval prowess led to a string of victories that helped lay bare the fragility of tsarist Russia's social, political, and economic fabric.

The Russian surrender of Port Arthur in January 1905 decisively gave Japan the upper hand. Over the war's course Japan's government borrowed tens of millions of pounds sterling in London, New York, and Berlin. In 1904, Japan had to pay an interest rate of around $7\frac{1}{2}$ percent per year on its borrowing, but by 1905, with the war's ultimate outcome no longer in doubt, lenders were charging Japan only around $5\frac{1}{2}$ percent. The Japanese neutralized Russia's naval forces in June 1905, and peace was concluded the following September.

The Russo-Japanese War offers an unusually good testing ground for the model we have developed: it caused no disruption of global financial markets and there was a fair amount of certainty as to the eventual winner. Figure 1.6 shows that Japan's current account moved sharply into deficit during the war, with foreign borrowing topping 10 percent of GDP in 1905. Also consistent with our model, national saving dropped sharply in the years 1904 and 1905, as shown in Table 1.2.

■

1.3.2 Saving and the Interest Rate

We now justify the shapes of the saving schedules drawn in Figure 1.5. This reasoning requires an understanding of the complex ways a change in the interest rate affects the lifetime consumption allocation.

1.3.2.1 The Elasticity of Intertemporal Substitution

The key concept elucidating the effects of interest rates on consumption and saving is the elasticity of intertemporal substitution, which measures the sensitivity of the intertemporal consumption allocation to an interest-rate change.

To see the role of intertemporal substitutability in determining the demands for consumption on different dates, take natural logarithms of the across-date first-order condition (4) and compute the total differential

$$\begin{aligned} \mathrm{d}\log(1+r) &= \frac{u''(C_1)}{u'(C_1)}\mathrm{d}C_1 - \frac{u''(C_2)}{u'(C_2)}\mathrm{d}C_2 \\ &= \frac{C_1u''(C_1)}{u'(C_1)}\mathrm{d}\log C_1 - \frac{C_2u''(C_2)}{u'(C_2)}\mathrm{d}\log C_2. \end{aligned} \tag{20}$$

Define the inverse of the elasticity of marginal utility by

$$\sigma(C) = -\frac{u'(C)}{Cu''(C)}. \tag{21}$$

The parameter defined in eq. (21) is called the *elasticity of intertemporal substitution.* When σ is constant, eq. (20) becomes

$$\mathrm{d}\log\left(\frac{C_2}{C_1}\right) = \sigma\,\mathrm{d}\log(1+r).$$

Intuitively, a gently curving period utility function (a high σ) implies a sensitive relative consumption response to an interest-rate change.

The class of period utility functions characterized by a constant elasticity of intertemporal substitution is

$$u(C) = \frac{C^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}, \qquad \sigma > 0. \tag{22}$$

We refer to this class of utility functions as the *isoelastic* class. For $\sigma = 1$, the right-hand side of eq. (22) is replaced by its limit, $\log(C)$.[14]

14. We really have to write the isoelastic utility function as

$$u(C) = \frac{C^{1-\frac{1}{\sigma}} - 1}{1-\frac{1}{\sigma}}$$

if we want it to converge to logarithmic as $\sigma \to 1$. To see convergence, we now can use L'Hospital's rule. As $\sigma \to 1$, the numerator and denominator of the function both approach 0. Therefore, we can differentiate both with respect to σ and get the answer by taking the limit of the derivatives' ratio, $C^{1-\frac{1}{\sigma}}\log(C)$, as $\sigma \to 1$.

Subtracting the constant $1/(1-\frac{1}{\sigma})$ from the period utility function does not alter economic behavior: the utility function in eq. (22) has exactly the same implications as the alternative function. To avoid burdening the notation, we will always write the isoelastic class as in eq. (22), leaving it implicit that one must subtract the appropriate constant to derive the $\sigma = 1$ case.

1.3.2.2 The Shape of the Saving Schedule

To determine the date 1 consumption response to an interest-rate change, use Home's intertemporal budget constraint, $C_2 = (1+r)(Y_1 - C_1) + Y_2$, to eliminate C_2 from its Euler equation, $u'(C_1) = (1+r)\beta u'(C_2)$. (We are assuming $B_1 = 0$.) The result is

$$u'(C_1) = (1+r)\beta u'\left[(1+r)(Y_1 - C_1) + Y_2\right].$$

Implicitly differentiating with respect to r gives

$$\frac{\mathrm{d}C_1}{\mathrm{d}r} = \frac{\beta u'(C_2) + \beta(1+r)u''(C_2)(Y_1 - C_1)}{u''(C_1) + \beta(1+r)^2u''(C_2)}. \tag{23}$$

Let's assume for simplicity that $u(C)$ is isoelastic with constant intertemporal substitution elasticity σ. We can then divide the numerator and denominator of the last equation by $u'(C_2)/C_2$ and, using definition (21) and the Euler equation (3), express the derivative as

$$\frac{\mathrm{d}C_1}{\mathrm{d}r} = \frac{(Y_1 - C_1) - \sigma C_2/(1+r)}{1+r+(C_2/C_1)}. \tag{24}$$

The numerator shows that a rise in r has an ambiguous effect on Home's date 1 consumption. The negative term proportional to σ represents substitution away from date 1 consumption that is entirely due to the rise in its relative price. But there is a second term, $Y_1 - C_1$, that captures the terms-of-trade effect on welfare of the interest rate change. If Home is a first-period borrower, $C_1 > Y_1$, the rise in the interest rate is a terms-of-trade deterioration that makes it poorer. As eq. (24) shows, this effect reinforces the pure relative-price effect in depressing C_1. But as r rises and Home switches from borrower to lender, the terms-of-trade effect reverses direction and begins to have a positive influence on C_1. For high enough interest rates, C_1 could even become an increasing function of r. If $Y_1 - C_1 > 0$, we can be sure that $\mathrm{d}C_1/\mathrm{d}r < 0$ only if r is not too far from the Home autarky rate.

Since date 1 output is given at Y_1, these results translate directly into conclusions about the response of saving S_1, which equals $Y_1 - C_1$. The result is a saving schedule **SS** such as the one in Figure 1.5. (Of course, the same principles govern an analysis of Foreign, from which the shape of **S*S*** follows.)

The possibility of a "perverse" saving response to the interest rate means that the world economy could have multiple equilibria, some of them unstable. Provided the response of total world saving to a rise in r is positive, however, the world market for savings will be stable (in the Walrasian sense), and the model's predictions still will be intuitively sensible. Introducing investment, as we do later, reduces the likelihood of unstable equilibria. Further analysis of stability is left for the chapter appendix. Our diagrammatic analysis assumes a unique stable world equilibrium.

1.3.2.3 The Substitution, Income, and Wealth Effects

A closer look at the consumption behavior implied by isoelastic utility leads to a more detailed understanding of how an interest rate change affects consumption.

Consider maximizing lifetime utility (1) subject to (2) when the period utility function is isoelastic. Since $u'(C) = C^{-1/\sigma}$ now, Euler equation (3) implies the dynamic consumption equation $C_1^{-1/\sigma} = (1+r)\beta C_2^{-1/\sigma}$. Raising both sides to the power $-\sigma$ yields

$$C_2 = (1+r)^{\sigma}\beta^{\sigma}C_1. \tag{25}$$

Using the budget constraint, we find that consumption in period 1 is

$$C_1 = \frac{1}{1+(1+r)^{\sigma-1}\beta^{\sigma}}\left(Y_1 + \frac{Y_2}{1+r}\right). \tag{26}$$

This consumption function reflects three distinct ways in which a change in the interest rate affects the individual:

1. *Substitution effect.* A rise in the interest rate makes saving more attractive and thereby induces people to reduce consumption today. Alternatively, a rise in r is a rise in the price of present consumption in terms of future consumption; other things the same, it should cause substitution toward future consumption.

2. *Income effect.* A rise in the interest rate also allows higher consumption in the future given the present value of lifetime resources. This expansion of the feasible consumption set is a positive income effect that leads people to raise present consumption and curtail their saving. The tension between this income effect and the substitution effect is reflected in the term $(1+r)^{\sigma-1}$ appearing in consumption eq. (26). When $\sigma > 1$ the substitution effect dominates because consumers are relatively willing to substitute consumption between periods. When $\sigma < 1$ the income effect wins out. When $\sigma = 1$ (the log case), the fraction of lifetime income spent on present consumption doesn't depend on the interest rate: it is simply $1/(1+\beta)$.

3. *Wealth effect.* The previous two effects refer to the fraction of lifetime income devoted to present consumption. The wealth effect, however, comes from the change in lifetime income caused by an interest rate change. A rise in r lowers lifetime income $Y_1 + Y_2/(1+r)$ (measured in date 1 consumption units) and thus reinforces the interest rate's substitution effect in lowering present consumption and raising saving.

As we have seen, the conflict among *substitution, income,* and *wealth effects* can be resolved in either direction: theory offers no definite prediction about how a change in interest rates will change consumption and saving. Section 1.3.4 will examine the interplay of these three effects in detail, for general preferences. A key conclusion of the analysis is that the income and wealth effects identified in this subsection add up to the *terms-of-trade effect* discussed in section 1.3.2.2's analysis of saving.

1.3.3 Investment, Saving, and the Metzler Diagram

We now introduce investment into the two-country model. Saving and investment can differ for an individual country that participates in the world capital market. In equilibrium, however, the world interest rate equates global saving to global investment. That equality underlies our adaptation of a classic diagram invented by Metzler (1960), updated to incorporate intertemporally maximizing decision makers.

1.3.3.1 Setup of the Model

Figure 1.7 graphs first-period saving and investment for Home and Foreign. Because we wish to study changes in investment productivity, let us now write the Home and Foreign production functions as

$$Y = AF(K), \qquad Y^* = A^*F^*(K^*),$$

where A and A^* are exogenously varying productivity coefficients. Home's investment curve (labeled **II**) traces out the analog of eq. (17), $A_2F'(K_1 + I_1) = r$ (where we remind you that K_1 is predetermined). Still marking Foreign symbols with asterisks, we write the corresponding equation for Foreign, which defines its investment curve (**I*I***), as $A_2^*F^{*\prime}(K_1^* + I_1^*) = r$. Because production functions are increasing but strictly concave, both investment curves slope downward.

Saving behavior appears somewhat trickier to summarize than in the endowment model of section 1.3.1. The reason is that investment now enters a country's budget constraint [recall eq. (15)], so interest rate effects on investment affect saving directly. To explore the saving schedules **SS** and **S*S*** we proceed as in the pure endowment case. Use Home's intertemporal budget constraint, which now is $C_2 = (1+r)[A_1F(K_1) - C_1 - I_1] + A_2F(K_1 + I_1) + K_1 + I_1$ [in analogy with the maximand (16)], to eliminate C_2 from its Euler equation, $u'(C_1) = (1+r)\beta u'(C_2)$. (We are assuming $B_1 = 0$.) The result is

$$u'(C_1) = (1+r)\beta u'\left\{(1+r)[A_1F(K_1) - C_1 - I_1] + A_2F(K_1 + I_1) + K_1 + I_1\right\}.$$

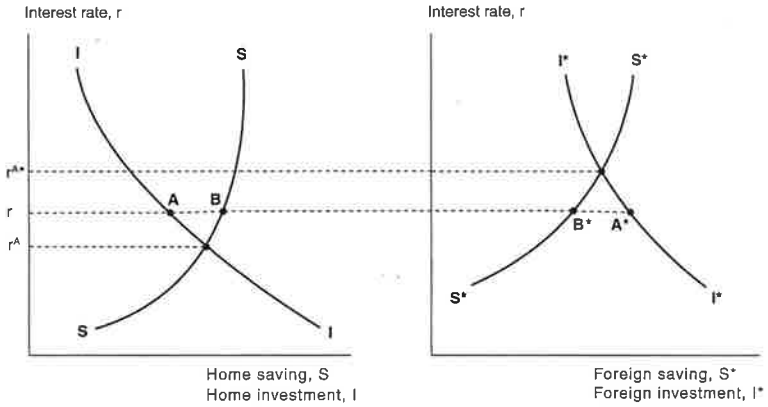


Figure 1.7
Global intertemporal equilibrium with investment

Implicitly differentiating with respect to r gives

$$\frac{dC_1}{dr} = \frac{\beta u'(C_2) + \beta(1+r)u''(C_2)\left\{[A_1F(K_1) - C_1 - I_1] + \left[A_2F'(K_1+I_1) - r\right]\frac{\partial I_1}{\partial r}\right\}}{u''(C_1) + \beta(1+r)^2u''(C_2)},$$

where $\partial I_1/\partial r$ represents the (negative) date 1 investment response to a rise in r. If we assume that saving decisions are based on profit-maximizing output and investment levels, as is natural, then the equality of the marginal product of capital and r, $A_2F'(K_1 + I_1) = r$, implies that the last derivative is precisely the same as eq. (23) in section 1.3.2.2, but with $Y_1 - C_1$ replaced by the date 1 current account for an investment economy with $B_1 = 0$, $A_1F(K_1) - C_1 - I_1$. For isoelastic utility, we have the analog of eq. (24),

$$\frac{dC_1}{dr} = \frac{(Y_1 - C_1 - I_1) - \sigma C_2/(1+r)}{1 + r + (C_2/C_1)},$$

which means that, given current account balances, the slopes of the saving schedules are the same as for pure endowment economies!

How can this be? The answer turns on a result from microeconomics that is useful at several points in this book, the *envelope theorem*.[15] The first-order condition for investment ensures that a small deviation from the optimum doesn't alter the present value of national output, evaluated at the world interest rate. When we compute the consumer's optimal response to a small interest rate change, it therefore doesn't matter whether production is being adjusted optimally: at the margin, the investment adjustment $\partial I_1/\partial r$ has no effect on net lifetime resources, and hence no effect on the consumption response.

15. For a description, see Simon and Blume (1994, p. 453).

Now consider the equilibrium in Figure 1.7. If Home and Foreign could not trade, each would have its own autarky interest rate equating country saving and investment. In Figure 1.7, Home's autarky rate, r^A, is below Foreign's, r^{A*}.

What is the equilibrium with intertemporal trade? Equilibrium requires

$$Y_1 + Y_1^* = C_1 + C_1^* + I_1 + I_1^*$$

(still omitting government spending). An alternative expression of this equilibrium condition uses eq. (13):

$$S_1 + S_1^* = I_1 + I_1^*.$$

The world as a whole is a closed economy, so it is in equilibrium when desired saving and investment are equal. Because $CA_1 = S_1 - I_1$, however [recall eq. (14)], the equilibrium world interest rate also ensures the mutual consistency of desired current accounts:

$$CA_1 + CA_1^* = 0.$$

The equilibrium occurs at a world interest rate r above r^A but below r^{A*}, as indicated by the equal lengths of segments **AB** and **B*A***. Home has a current account surplus in period 1, and Foreign has a deficit, in line with comparative advantage. The equilibrium resource allocation is *Pareto optimal*, or efficient in the sense that there is no way to make everyone in the world economy better off. (This was also true in the pure endowment case, of course.) Since both countries face the same world interest rate, their intertemporal optimality conditions (4) imply equal marginal rates of substitution of present for future consumption. The international allocation of capital also is efficient, in the sense that capital's date 2 marginal product is the same everywhere: $A_2F'(K_2) = A_2^*F^{*\prime}(K_2^*) = r$.

1.3.3.2 Nonseparation of Investment from Saving

Having derived a Metzler diagram, we are ready for applications. Consider first an increase in Home's impatience, represented by a fall in the parameter β. In Figure 1.7 this change would shift Home's saving schedule to the left, raising the equilibrium world interest rate and reducing Home's date 1 lending to Foreign.

Notice that investment falls everywhere as a result. Unlike in the small country case, a shift in a large country's consumption preferences can affect investment by moving the world interest rate. In the present example, saving and investment

move in the same direction (down) in Home but in opposite directions (saving up, investment down) in Foreign.

1.3.3.3 Effects of Productivity Shifts

In section 1.3.1 we asked how exogenous output shocks affect the global equilibrium. For small changes, the envelope theorem implies that the saving curves here will respond to shifts in the productivity factors A_2 and A_2^* as if these were purely exogenous output changes not warranting changes in investment. (Shifts in A_1 and A_1^* plainly are of this character because K_1 can't be adjusted retroactively!) Thus shifts in productivity affect **SS** and **S*S*** precisely as the corresponding output shifts in section 1.3.1 did.

The investment schedules, however, also shift when capital's future productivity changes. At a constant interest rate, the (horizontal) shift in **II** due to a rise in A_2 comes from differentiating the condition $A_2F'(K_1 + I_1) = r$:

$$\left.\frac{dI_1}{dA_2}\right|_{r \text{ constant}} = -\frac{F'(K_2)}{A_2F''(K_2)} > 0.$$

A rise in A_2^* has a parallel effect on **I*I***.

Let's use the model first to consider a rise in date 1 Home productivity A_1. As in the endowment model, Home saving increases at every interest rate. Thus **SS** in Figure 1.7 shifts to the right, pushing the world interest rate down, as before. What is new is the response of investment, which rises in both countries. Home's date 1 current account surplus rises, as does Foreign's deficit.

Next think about a rise in A_2, which makes Home's capital more productive in the second period. In Figure 1.8, which assumes zero current accounts initially, Home's investment schedule shifts to the right. At the same time, Home's saving schedule shifts to the left because future output is higher while first-period output is unchanged. The world interest rate is unambiguously higher. Since Foreign's curves haven't shifted, its saving is higher and its investment is lower. The result is a current account surplus for Foreign and a deficit for Home.

In Figure 1.8 the ratio of Home to Foreign investment is higher as a result of A_2 rising, but since the level of Foreign investment is lower, it isn't obvious that Home investment actually rises. Perhaps surprisingly, it is theoretically possible for Home investment to *fall* because of the higher world interest rate. Predictions as to what might occur in practice therefore must rest on empirical estimates of preference parameters and production functions. The next application illustrates the underlying reasoning by considering the related question of how total world investment responds to a change in expected future productivity.

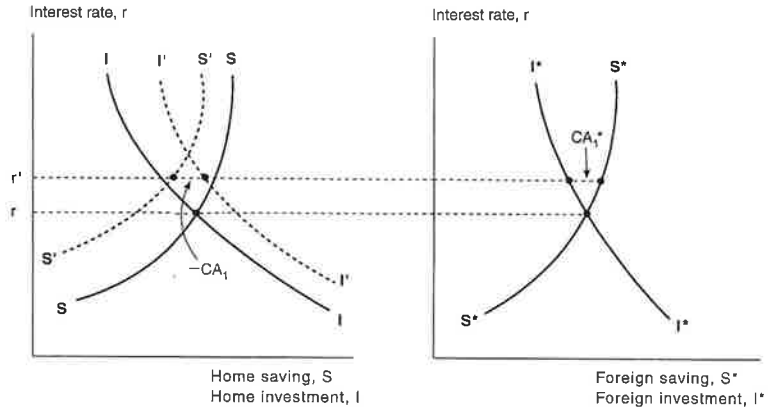


Figure 1.8
A rise in future Home productivity

Application: Investment Productivity and World Real Interest Rates in the 1980s

In the early 1980s world real interest rates suddenly rose to historically high levels, sparking a lively debate over the possible causes. Figure 1.9 shows a measure of global real interest rates since 1960.[16]

An influential paper by Blanchard and Summers (1984) identified anticipated future investment profitability as a prime explanatory factor. In support of their view, they offered econometric equations for the main industrial countries showing that investment in 1983 and early 1984 was higher than one would have predicted on the basis of factors other than the expected future productivity of capital.

Subsequent empirical work by Barro and Sala-i-Martin (1990) showed that a stock-market price proxy for expected investment profitability had a positive effect on both world investment and the world real interest rate. This evidence, suggesting that a rise in expected investment profitability could indeed have caused the world-wide increase in investment and interest rates observed in 1983–84, lent retroactive support to the Blanchard-Summers thesis.

The issue is easily explored in our global equilibrium model. Since the main points do not depend on differences between Home and Foreign, it is simplest to assume that the two countries are completely identical. In this case, one can think

16. The data shown are GDP-weighted averages of ten OECD countries' annual average real interest rates. Real interest rates are defined as nominal long-term government bond rates less actual consumer-price index inflation over the following year.

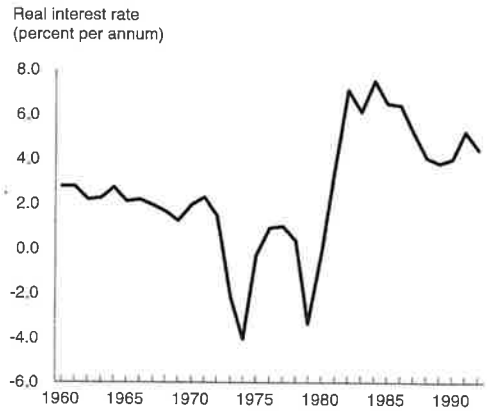


Figure 1.9
World ex post real interest rates, 1960–92

of the world as a single closed economy populated by a single representative agent and producing output according to the single production function $Y = AK^{\alpha}$, $\alpha < 1$. Consider now the effects of an increase in the date 2 productivity parameter A_2 that characterizes Home and Foreign industry alike. Figure 1.10 supplies an alternative way to picture global equilibrium. It shows that the productivity disturbance raises world investment demand, which we continue to denote by $I_1 + I_1^*$, at every interest rate. We also know, however, that a rise in future productivity lowers world saving, $S_2 + S_2^*$, at every interest rate. The move from equilibrium **A** to equilibrium **B** in the figure appears to have an ambiguous effect on world investment.

To reach a more definite answer, we compare the vertical distances by which the two curves shift. A proof that the world saving schedule shifts further upward than the world investment schedule is also a proof that world investment must fall.

As a first step, we compute the shift in the investment curve. For the production function we have assumed, the optimal second-period capital stock is $K_2 = (A_2\alpha/r)^{1/(1-\alpha)}$, as end-of-chapter exercise 3 asks you to show. As a result, the world investment schedule is defined by

$$I_1 + I_1^* = (A_2\alpha/r)^{1/(1-\alpha)} - K_1.$$

The vertical shift induced by a rise dA_2 in A_2 is the change in r, dr, consistent with $d(I + I^*) = 0$. Since world investment clearly remains constant if r rises precisely in proportion to A_2,

$$dr|_{I+I^* \text{ constant}} = r\frac{dA_2}{A_2} = r\hat{A}_2,$$

where a "hat" denotes a small percentage change.

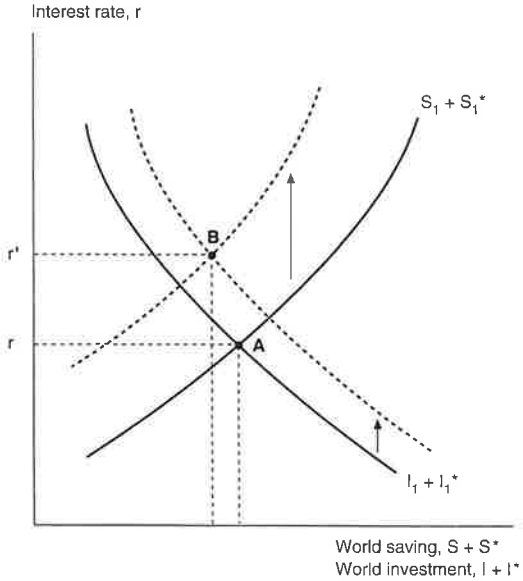


Figure 1.10
A rise in world investment productivity

To compare this shift with that of the world saving schedule, let's assume temporarily that Home and Foreign residents share the logarithmic lifetime utility function $U_1 = \log C_1 + \beta \log C_2$. End-of-chapter exercise 3 implies that date 1 world saving can be expressed as

$$S_1 + S_1^* = \frac{\beta}{1+\beta}A_1F(K_1) + \frac{1}{1+\beta}\left[K_2 - K_1 - \frac{A_2F(K_2) + K_2}{1+r}\right].$$

Differentiate this schedule with respect to r and A_2, imposing $d(S_1 + S_1^*) = 0$. The envelope theorem permits omission of the induced changes in K_2, so the result of differentiation is

$$\frac{-1}{1+\beta}\left[\frac{F(K_2)}{1+r}dA_2 - \frac{A_2F(K_2) + K_2}{(1+r)^2}dr\right]$$

$$= \frac{-1}{1+\beta}\left[\frac{A_2F(K_2)}{1+r}\hat{A}_2 - \frac{A_2F(K_2) + K_2}{(1+r)^2}dr\right]$$

$$= d(S_1 + S_1^*) = 0.$$

The solution for dr is

$$dr|_{S+S^* \text{ constant}} = (1+r)\frac{A_2F(K_2)}{A_2F(K_2)+K_2}\hat{A}_2$$

$$= (1+r)\frac{A_2^{\frac{1}{1-\alpha}}\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}}}{A_2^{\frac{1}{1-\alpha}}\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}} + A_2^{\frac{1}{1-\alpha}}\left(\frac{\alpha}{r}\right)^{\frac{1}{1-\alpha}}}\hat{A}_2$$

$$= \frac{1+r}{1+\frac{\alpha}{r}}\hat{A}_2 > r\hat{A}_2 = dr|_{I+I^* \text{ constant}}.$$

Thus, as Figure 1.10 shows, world investment really is lower at the new equilibrium **B** than at **A**.

The preceding result obviously flows from assuming logarithmic preferences, which imply an intertemporal substitution elasticity $\sigma = 1$. How would the outcome in Figure 1.10 be affected by making σ smaller? Making σ smaller would only make it more likely that rises in future investment productivity push world investment down. The factor driving the seemingly perverse result of the log-utility case is a wealth effect: people want to spread the increase in period 2 income over both periods of life, so they reduce period 1 saving, pushing the real interest rate so high that investment actually falls. But if $\sigma < 1$, the desire for smooth consumption is even stronger than in the log case. Thus the interest rate rises and investment falls even more sharply.

We conclude that while an increase in expected investment profitability can in principle explain a simultaneous rise in real interest rates and current investment, as Blanchard and Summers (1984) argued, this outcome is unlikely unless individuals have relatively high intertemporal substitution elasticities. Economists disagree about the likely value of σ, but while there are many estimates below 0.5, few are much higher than 1. We are left with a puzzle. Without positing a rise in expected investment profitability, it is hard to explain the comovement of investment and real interest rates in 1983–84. But if the consensus range of estimates for σ is correct, this change probably should have lowered, not raised, world investment.

Can the simultaneous rise in investment and real interest rates be explained if both *current* capital productivity *and* future profitability rose together? Under that scenario, the fall in saving is reduced, but so is the accompanying rise in the interest rate. This would only leave a greater portion of the sharp increase in real interest rates unexplained.

The empirical record would seem to bear out our theoretical skepticism of the view that expected future productivity growth caused the high real interest rates of the early 1980s. World investment actually turned out to be lower on average after

1983. The fact that *future* investment (as opposed to current investment) dropped suggests that, ex post, productivity did not rise. We will look at the real interest rate puzzle again from the angle of saving at the end of the chapter. ■

*1.3.4 Real Interest Rates and Consumption in Detail

This section examines the substitution, income, and wealth effects of section 1.3.2.3 in greater detail. As a notational convenience, we define the market discount factor

$$R \equiv \frac{1}{1+r}$$

as the price of future consumption in terms of present consumption. We simplify by holding $G_1 = G_2 = 0$ and assuming exogenous output on both dates.

1.3.4.1 The Expenditure Function and Hicksian Demands

The easiest way to understand substitution, income, and wealth effects is to use the *expenditure function*, denoted by $E(R, U_1)$. It gives the minimal lifetime expenditure, measured in date 1 output, that enables a consumer to attain utility level U_1 when the price of future consumption is R. In Chapter 4 we will use the expenditure function to construct price indexes.[17]

We will need one main result on expenditure functions. Define the *Hicksian consumption demands* as the consumption levels a consumer chooses on the two dates when lifetime utility is U_1. We write the Hicksian demands as $C_1^H(R, U_1)$ and $C_2^H(R, U_1)$. The result we need states that the partial derivative of the expenditure function with respect to R is the Hicksian demand for date 2 consumption:[18]

$$C_2^H(R, U_1) = E_R(R, U_1). \tag{27}$$

This result and the budget constraint imply that $C_1^H(R, U_1) = E(R, U_1) - RE_R(R, U_1)$.

17. Dixit and Norman (1980) provide the classic treatise on the use of expenditure functions in static international trade theory.

18. The Hicksian demands satisfy $C_1^H(R, U_1) + RC_2^H(R, U_1) = E(R, U_1)$. Differentiating partially with respect to R gives us

$$\frac{\partial C_1^H}{\partial R} + R\frac{\partial C_2^H}{\partial R} + C_2^H = E_R.$$

These partial derivatives are taken with the utility level U_1 held constant. But along a fixed utility curve, the ratio $\frac{\partial C_1^H}{\partial R} / \frac{\partial C_2^H}{\partial R}$ is just minus the marginal rate of substitution of C_1^H for C_2^H, which equals $-R$. The equality $E_R = C_2^H$ follows. (This is another example of the envelope theorem.)

1.3.4.2 Income and Wealth Effects

The income and wealth effects of a change in R reflect its impact on lifetime utility, U_1. The expenditure function yields a slick derivation of this impact. The equilibrium lifetime utility level of a maximizing representative individual is given implicitly by

$$E(R, U_1) = Y_1 + RY_2.$$

Totally differentiating with respect to R gives us

$$E_R dR + E_U dU_1 = Y_2 dR,$$

which, using eq. (27), can be solved to yield the income-cum-wealth effect

$$E_U \frac{dU_1}{dR} = Y_2 - C_2^H(R, U_1) = Y_2 - C_2. \tag{28}$$

This equation formalizes the basic intuition about terms-of-trade effects mentioned in section 1.3.1. When $Y_2 > C_2$, a country is repaying past debts incurred through a current account deficit on date 1. A rise in the price of future consumption, R, is a fall in the interest rate, r, and an improvement in the country's terms of intertemporal trade. Thus a rise in R has a positive welfare effect in this case, but a negative effect when the country is an importer of future consumption.

1.3.4.3 Substitution Effect and Slutsky Decomposition

Hicksian demand functions are useful for a decomposition of interest-rate effects because their price derivatives show the pure substitution effects of price changes, that is, the effects of price changes after one controls for income and wealth effects by holding the utility level constant. In this chapter, however, we have focused on *Marshallian* demand functions that depend on wealth rather than utility. Define wealth on date 1, W_1, as the present value of the consumer's lifetime earnings:

$$W_1 \equiv Y_1 + RY_2.$$

Then eq. (26), for example, implies the Marshallian demand function for date 1 consumption,

$$C_1(R, W_1) = \frac{W_1}{1 + \beta^\sigma R^{1-\sigma}} = \frac{Y_1 + RY_2}{1 + \beta^\sigma R^{1-\sigma}},$$

which expresses date 1 consumption demand as a function of the interest rate and wealth.

As we now show, the total derivative dC_1/dR of this Marshallian demand is the sum of a Hicksian substitution effect, an income effect, *and* a wealth effect.

The proof relies on an important identity linking Marshallian and Hicksian demands: the Marshallian consumption level, given the minimum lifetime expenditure needed to reach utility U_1, equals its Hicksian counterpart, given U_1 itself. Thus, for date 1 consumption,

$$C_1\left[R, E(R, U_1)\right] = C_1^H(R, U_1). \tag{29}$$

With the machinery we have now developed, it is simple to show how substitution, income, and wealth effects together determine the response of consumption and saving to interest-rate changes. Partially differentiate identity (29) with respect to R (holding U_1 constant) and use eq. (27). The result is the famous Slutsky decomposition of partial price effects,

$$\frac{\partial C_1(R, W_1)}{\partial R} = \frac{\partial C_1^H(R, U_1)}{\partial R} - \frac{\partial C_1}{\partial W_1} C_2. \tag{30}$$

The two terms on the right-hand side of this equation are, respectively, the substitution effect and the income effect.

The interest-rate effect analyzed in section 1.3.2.3 was, however, the *total* derivative of C_1 with respect to R. The total derivative is, using eq. (30) and the wealth effect $dW_1/dR = Y_2$,

$$\begin{aligned}\frac{dC_1(R, W_1)}{dR} &= \frac{\partial C_1(R, W_1)}{\partial R} + \frac{\partial C_1(R, W_1)}{\partial W_1}\frac{dW_1}{dR} \\ &= \frac{\partial C_1^H(R, U_1)}{\partial R} + \frac{\partial C_1(R, W_1)}{\partial W_1}(Y_2 - C_2).\end{aligned} \tag{31}$$

This equation shows that the total effect of the interest rate on present consumption is the sum of the pure substitution effect and a term that subtracts the income from the wealth effect. Looking again at eq. (28), we see that the latter difference is none other than the consumption effect of a change in wealth equivalent to the intertemporal terms-of-trade effect. Together, the income and wealth effects push toward a rise in consumption on both dates and a fall in saving for a country whose terms of intertemporal trade improve, and the opposite effects for one whose terms of intertemporal trade worsen.

1.3.4.4 The Isoelastic Intertemporally Additive Case

All the results in this subsection have been derived without reference to a specific utility function: we have not even assumed intertemporal additivity. When the lifetime utility function is additive and the period utility functions are isoelastic [recall eq. (22)], closed-form solutions for the Hicksian demands can be derived. For example, it is a good homework problem to show that the Hicksian demand function for date 1 consumption is

$$C_1^H(R, U_1) = \left[\frac{\left(1 - \frac{1}{\sigma}\right) U_1}{1 + \beta^\sigma R^{1-\sigma}} \right]^{\frac{\sigma}{\sigma-1}}.$$

You can also show that decomposition (31) takes the form

$$\frac{dC_1}{dR} = \sigma \frac{\beta^\sigma R^{-\sigma}}{1 + \beta^\sigma R^{1-\sigma}} C_1 + \frac{1}{1 + \beta^\sigma R^{1-\sigma}} (Y_2 - C_2).$$

Here, the first term on the right is the pure substitution effect and the second is the difference between the wealth and income effects. Applying the Euler equation (25) to this expression transforms the derivative into

$$\frac{dC_1}{dR} = (\sigma - 1) \frac{\beta^\sigma R^{-\sigma}}{1 + \beta^\sigma R^{1-\sigma}} C_1 + \frac{1}{1 + \beta^\sigma R^{1-\sigma}} Y_2.$$

This version makes apparent that the sign of $\sigma - 1$ determines whether the substitution or income effect is stronger. You can verify that the last equation is equivalent to eq. (24) derived in section 1.3.2.2.

1.4 Taxation of Foreign Borrowing and Lending

Governments sometimes restrict international borrowing and lending by taxing them. In this section we look at the taxation of international capital flows in a two-country world and produce a possible nationalistic rationale for taxation.

Government intervention in the international loan market potentially can raise *national* welfare while reducing that of trading partners and pushing the world economy as a whole to an inefficient resource allocation. The mechanism is the one at work in the classic "optimal tariff" argument in trade theory: through taxation, a government can exploit any collective monopoly power the country has to improve its intertemporal terms of trade and, thereby, to raise national welfare.

For simplicity we return to the pure endowment case with logarithmic utility, which is the subject of a detailed end-of-chapter exercise. Suppose initially that Home is a command economy in which the government chooses C_1 and C_2. Both Home and Foreign are large enough to influence world prices, but individual Foreign actors continue to be competitive price takers. There is therefore a supply schedule for Foreign savings,

$$S_1^*(r) = Y_1^* - C_1^*(r) = \frac{\beta^*}{1 + \beta^*} Y_1^* - \frac{1}{(1 + \beta^*)(1 + r)} Y_2^*.$$

Home's government sees matters differently from competitive individuals. It knows that changes in its consumption choices affect the world interest rate and that the world interest rate is determined by the equilibrium condition $Y_1 + Y_1^* = C_1 + C_1^*(r)$. By combining this condition with the Foreign saving schedule, Home's government can calculate how the world interest rate rises as C_1 rises,

$$1 + r = \frac{Y_2^*}{(1 + \beta^*)(Y_1 - C_1) + \beta^* Y_1^*}.$$

Putting this equation and the Home budget constraint, $C_2 = Y_2 - (1 + r)(C_1 - Y_1)$, together, we get the trade-off between present and future consumption as perceived by Home's government,

$$C_2 = Y_2 + \frac{Y_2^*}{(1 + \beta^*)(Y_1 - C_1) + \beta^* Y_1^*} (Y_1 - C_1). \tag{32}$$

This trade-off describes, of course, the Foreign *offer curve* of trade theory.

Figure 1.11 illustrates Home's position in the case $r^A > r^{A*}$. The heavier straight line passing through the endowment point **A** has slope $-(1 + r^L)$, where r^L is the equilibrium world interest rate that would prevail were both governments to follow laissez-faire principles and allow free trade. The curve **TT** passing through **A** is the graph of eq. (32); it shows the consumption possibilities open to Home when Foreign residents are price takers while Home's government sets domestic consumption taking into account its influence on the world interest rate.

The key feature to notice about Figure 1.11 is that part of **TT** lies *strictly outside* the laissez-faire budget line. How can we be sure this is the case?

Differentiation of eq. (32) shows that the slope of **TT** at **A** is

$$\left. \frac{dC_2}{dC_1} \right|_{C_1 = Y_1} = -\frac{Y_2^*}{\beta Y_1^*} = -(1 + r^{A*})$$

[recall eq. (7), which parallels the second equality in this expression]. The interpretation of this derivative is intuitive. At their autarky consumption allocation, Foreign residents are willing to lend or borrow a small amount at their autarky interest rate, r^{A*}, which we know is below the laissez-faire equilibrium rate, r^L. Since Home's laissez-faire consumption choice at **B** certainly remains feasible when Home's government internalizes the country's market power, point **B** lies on **TT**, and that locus therefore has the shape shown in Figure 1.11.

Point **B** is not the preferred consumption point of a Home planner bent on maximizing national welfare. The planner would instead pick point **C**, which is feasible and yields higher national utility.

In a decentralized economy, the Home government can use a tax on foreign borrowing to induce price-taking residents to choose point **C** on their own. Notice first that if **C** is to be chosen by price-taking individuals, the slope of their indifference curves at **C** must equal $-(1 + r^T + \tau)$, where τ is the tax an individual pays the government for each unit of output borrowed from abroad and r^T is the equilibrium world interest rate when the tax is in place.

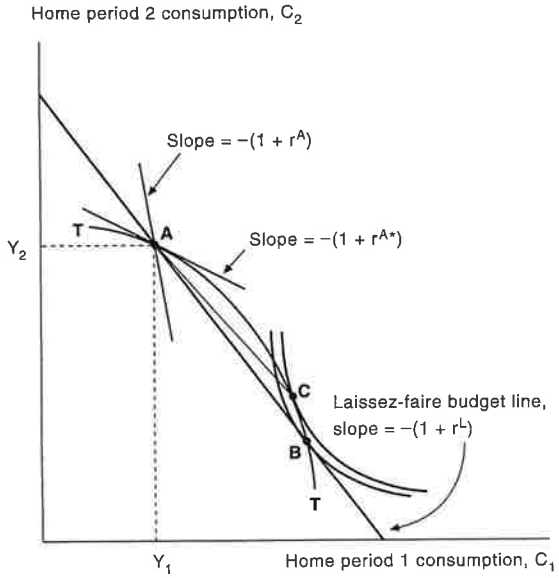


Figure 1.11
The optimal tax on foreign borrowing

We can read r^τ off of Figure 1.11 by noting that the economy's trade must be intertemporally balanced at the world rate of interest, that is, $Y_2 - C_2 = (1 + r^\tau)(C_1 - Y_1)$. (We are assuming here that the government rebates tax proceeds to its citizens in the form of lump-sum transfer payments.) Because points **A** and **C** both are feasible given the borrowing tax, this budget constraint implies that $-(1 + r^\tau)$ must equal the slope of the segment **AC**. Therefore, the optimal tariff τ is simply the wedge between the slope of **AC** and the slope of the indifference curve passing through **C**.[19]

Methods other than taxation can similarly enable the government to maximize Home's welfare. Most simply, the government could impose a quota to limit residents' borrowing to the amount corresponding to point **C** in Figure 1.11.

Observe that government policy drives the world interest rate below its laissez-faire level: $r^\tau < r^L$. Thus the Home government's exercise of monopoly power drives down the interest rate Foreign earns on its loans to Home: Home's intertemporal terms of trade improve. Correspondingly, Foreign is impoverished. Home's

19. As a test of understanding, reproduce the argument in the text for the case $r^A < r^{A*}$, and show that the optimal policy is a tax on foreign lending.

advantage comes from appropriating some of Foreign's potential gains from trade. Home's borrowing tax thus is a beggar-thy-neighbor policy, one that benefits a country at its trading partners' expense. Although Home also forgoes some trade gains, this loss is more than offset by the better borrowing terms its tax artificially creates.

Finally, Home's policy, by reducing the volume of intertemporal trade, moves the world as a whole away from an efficient, or Pareto-optimal, resource allocation. Because of this global inefficiency, Foreign's government could, in principle, bribe Home's not to impose the tax, while still leaving its own residents better off than under the tax.

Alternatively, Foreign's government, which we have assumed passively to follow a laissez-faire policy, could retaliate by imposing a tax on international lending. How would the world economy fare if the two governments ended up having a trade war? Alternatively, could they reach a negotiated solution that avoids such a confrontation? These possibilities are at the heart of everyday conflicts over trade and macroeconomic policies, an area we revisit in Chapter 9.

Notice that a *small* country, one whose output and consumption are dwarfed by foreign output, has no scope to influence the trade-off schedule (32): it faces the rest-of-world gross interest rate $Y_2^*/\beta^* Y_1^*$ no matter what it does. For a small country there is thus no terms-of-trade gain to offset the tax's distorting effect on trade. As a result, a small country's optimal tax on foreign borrowing is zero.

1.5 International Labor Movements

A central assumption in most models of international trade and finance is that labor is much less mobile internationally than either commodities or capital. Language and cultural barriers, family and ethnic ties, and political barriers all work to make international migration difficult. And, with only a few major exceptions such as Australia, Canada, and the United States, industrial countries have experienced low levels of immigration over the past twenty years. Table 1.3's figures for 1974–87 illustrate the magnitudes of industrial countries' recent net inflows of foreign workers.

For developing countries and countries of the former Soviet bloc, however, outward migration of workers is substantial. Even in more developed countries like Greece, where workers' remittances from abroad were 3 percent of GDP in 1992, international labor mobility has important economic consequences. Up until World War I, footloose labor flowed in massive waves from Europe to former European colonies in North America and Oceania (again see Table 1.3), as well as to Latin America. As recently as 1950–73, Germany and Switzerland let in numerous "guest workers" from Mediterranean countries, while France welcomed

Table 1.3
Industrial Country Net Immigration (average per year, as a percent of labor force)

Country	1870–1913	1914–49	1950–73	1974–87
Australia	0.96	0.74	2.06	1.14
Belgium	0.15	0.16	0.34	−0.02
Canada	1.08	0.15	1.35	0.50
France	0.11	−0.03	0.75	0.11
Germany	−0.48	−0.08	1.12	0.26
Italy	−0.64	−0.25	−0.41	0.17
Japan	n.a.	0.02	−0.01	−0.02
Netherlands	−0.18	−0.03	0.04	−0.02
Norway	−1.63	−0.33	0.00	0.30
Sweden	−0.92	0.06	0.38	0.28
Switzerland	0.07	−0.15	1.15	0.00
United Kingdom	−0.97	−0.24	−0.10	0.00
United States	1.38	0.35	0.47	0.51

Source: Maddison (1991) and OECD, *Labor Force Statistics*. The figures were calculated by dividing Maddison's average net immigration series by labor force membership as of 1890, 1929, 1960, and 1981.

immigrants from its former African colonies. These experiences justify a close examination of the causes and effects of international labor movements.

Though immigration usually is a much more emotion-charged issue than capital mobility is, labor mobility can yield similar efficiency benefits by equalizing marginal products of labor across countries.[20] To illustrate this point we develop a two-period small-country model in which there is no international borrowing or lending (perhaps the result of prohibitive barriers to international capital movements). International emigration and immigration are, however, completely free. How should we think about the gains from trade in such a world?

1.5.1 Capital, Labor, and the Production Function: A Digression

Since an analysis of labor flows must account for the impact of labor on output, we now make explicit that produced output is a function of capital *and* labor inputs,

$$Y = F(K, L). \tag{33}$$

The production function $F(K, L)$ shows *constant returns to scale* in the two factors; that is, for any number ξ, $F(\xi K, \xi L) = \xi F(K, L)$. Some properties of constant-returns (also known as linear homogeneous) production functions figure

20. With internationally identical production technologies, labor and capital marginal products could be equalized even without factor mobility across borders. This factor-price equalization can occur in a multigood model with free and costless trade, along the lines of the classic Heckscher-Ohlin model (see Dixit and Norman, 1980). All empirical evidence suggests that even when capital is somewhat mobile, equalization of real wages applies, if at all, only to the very long run.

prominently in the present model and in later chapters. We digress briefly to remind you of them.

First, output equals the sum of factor marginal products multiplied by factor inputs:

$$Y = F(K, L) = F_K(K, L)K + F_L(K, L)L. \tag{34}$$

Second, the marginal products of capital and labor depend only on the capital-labor ratio, $k \equiv K/L$. Because $F(K, L) = LF(K/L, 1)$,

$$F_K(K, L) = f'(k), \tag{35}$$

where $f(k) \equiv F(K/L, 1)$ is the "intensive" or per-worker production function.[21] From these two results we derive a third,

$$F_L(K, L) = f(k) - f'(k)k. \tag{36}$$

1.5.2 A Two-Period, Small-Country Model

Now we can get on with the model. On date 2 (the model's second period), the home economy's output is

$$Y_2 = F(K_2, L_2),$$

where K_2 is the capital that domestic residents accumulate during period 1. For simplicity, we assume that output on date 1, Y_1, is exogenous, and that the starting capital stock, K_1, is zero. Thus production takes place only on date 2.

Our small country faces a given world wage rate, w, at which it can export or import labor services on date 2. The representative resident has an inelastically supplied labor endowment of L^H on date 2, but the economy employs a total of L_2 labor units, where L_2 can be greater or less than L^H as a result of trade in labor services.[22] There is no government consumption. Because neither international borrowing nor lending is possible, the representative home-country individual maximizes $U_1 = u(C_1) + \beta u(C_2)$ subject to the constraints

21. Totally differentiate the equation $F(\xi K, \xi L) = \xi F(K, L)$ with respect to ξ and evaluate the result at $\xi = 1$; the implication is $F(K, L) = F_K(K, L)K + F_L(K, L)L$, a special case of Euler's theorem on homogeneous functions. To derive the second result mentioned in the text, observe that

$$F_K(K, L) = \lim_{\Delta K \to 0} \frac{F(K + \Delta K, L) - F(K, L)}{\Delta K}$$

$$= \lim_{\Delta K \to 0} \frac{F[k + (\Delta K)/L, 1] - F(k, 1)}{(\Delta K)/L} = f'(k).$$

22. If $L^H > L_2$, one can think of the representative individual as working abroad part-time or, equivalently, of some domestic workers emigrating to work abroad.

$$C_1 = Y_1 - K_2,$$

$$C_2 = L_2 f(K_2/L_2) - w(L_2 - L^{\text{H}}) + K_2.$$

The first of these constraints is self-explanatory; it differs from the first-period constraint under capital mobility in that borrowing from abroad can't be used to supplement first-period resources.[23] The second constraint states that date 2 consumption equals domestic product (a function of the economy's total employment), less the net wage payments on imported labor services, less investment (where we remind you that, as explained in section 1.2.2, $I_2 = -K_2$).

Using the two constraints to eliminate C_1 and C_2 from the utility function, we can write the representative individual's problem as

$$\max_{K_2, L_2} u(Y_1 - K_2) + \beta u\left[L_2 f(K_2/L_2) - w(L_2 - L^{\text{H}}) + K_2\right].$$

First-order conditions with respect to K_2 and L_2 are

$$u'(C_1) = \beta[1 + f'(k_2)]u'(C_2), \tag{37}$$

$$w = f(k_2) - f'(k_2)k_2, \tag{38}$$

where we have defined $k_2 \equiv K_2/L_2$ as the capital-labor ratio in production during period 2. The first of these equations is the Euler equation when the domestic real interest rate equals the marginal product of capital [recall eq. (35)]. The second states that under free international labor mobility, the marginal product of labor [recall eq. (36)] must equal the world wage rate.

Notice that eq. (38) ties down $k_2 = K_2/L_2$ as a function of the world wage rate, so we can write the production capital-labor ratio as a function $k(w)$, with $k'(w) = -1/k(w)f''[k(w)] > 0$. By implication, the domestic interest rate also is a function of w alone: it is given by $r(w) = f'[k(w)]$, so that $r'(w) = f''[k(w)]k'(w) = -1/k(w) < 0$. The model's negatively sloped functional relation between r and w is called the *factor-price frontier*. A rise in the world wage raises the optimal capital intensity of production, lowering the domestic marginal product of capital. As long as w doesn't change, however, neither k nor r can change, no matter what else in the economy *does* change (other than the production function itself). Given w, net labor exports adjust to ensure that k and r remain constant.

1.5.3 Pattern of Trade and Gains from Trade

Figure 1.12 illustrates the economy's equilibrium in a way that parallels our earlier discussion of capital mobility. The concave locus is the autarky PPF, which describes the home economy's intertemporal production (and consumption) tradeoff when the labor used in second-period production is restricted to the domestic

23. To ensure that the nonnegativity constraint on capital doesn't bind, the appropriate condition on the production function is now $\lim_{K \to 0} F_K(K, L) = \infty$.

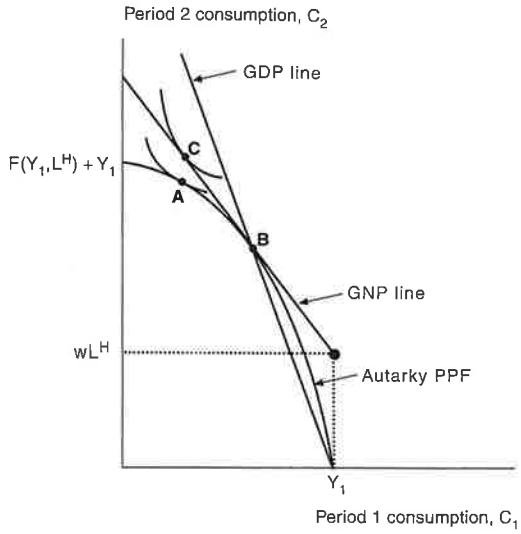


Figure 1.12
Trade in labor services

endowment, L^{H}. The autarky PPF is described by

$$C_2 = F\left(Y_1 - C_1, L^{\text{H}}\right) + Y_1 - C_1.$$

Its slope is (minus) $1 + F_K(K_2, L^{\text{H}})$, its horizontal intercept is at $C_1 = Y_1$, and its vertical intercept is at $C_2 = F(Y_1, L^{\text{H}}) + Y_1$. Point **A** is the economy's autarky equilibrium.

The straight line tangent to the autarky PPF at **B** is the representative domestic resident's intertemporal budget constraint when there is trade in labor services. It is described by the equation

$$C_2 = [1 + r(w)](Y_1 - C_1) + wL^{\text{H}},$$

and its linearity is due to the constancy of $r(w)$, given w. We call this line the *GNP line* because it equates second-period consumption plus investment to domestic output plus net factor payments from abroad. Point **B** is generated by an investment level at which the marginal product of capital is $r(w)$ when employment L_2 equals L^{H}.

The economy does not have to consume and produce at point **B**, however. By importing labor services from abroad [and doing enough extra investment to maintain $f'(k_2) = r(w)$], residents can move consumption up and to the left along the GNP line from **B**; by exporting labor services (and investing less) they can move down

and to the right. Because K_2 can't be less than zero (and by assumption there is no foreign borrowing), C_2 can never be less than wL^H. At this lowest point on the GNP line, the country consumes all its first-period output, has a domestic product of zero in the second period, and exports all its labor services abroad to generate a GNP of wL^H.

In Figure 1.12 the economy's preferred consumption point is **C**. At **C**, the economy is investing more than it would at **B**. To hold the marginal product of capital at $r(w)$, the economy imports labor services from abroad, raising L_2 above L^H.

Also passing through point **B** on the autarky PPF is the *GDP line*, which shows how the sum of second-period GDP and K_2 changes with first-period consumption decisions. (Remember that K_2 is eaten after date 2 production.) Recalling eq. (34) and noting that $L_2 = K_2/k(w)$, we see that the GDP line is described by

$$Y_2 + K_2 = F(K_2, L_2) + K_2 = [1 + r(w)]K_2 + wL_2$$
$$= \left[1 + r(w) + \frac{w}{k(w)}\right](Y_1 - C_1).$$

The GDP line passes through point **B** because there, net labor imports are zero and GDP = GNP. The GDP line is steeper than the GNP line, however. As investment rises, so does immigration, and net payments to foreigners place a wedge between GDP and GNP. Thus GDP exceeds GNP above **B** but is less than GNP below **B**.

At the autarky point **A** in Figure 1.12, the marginal product of capital is less than $r(w)$. The factor-price frontier therefore implies that the autarky wage rate, w^A, is greater than the world wage, w. It is straightforward to show that any country with $w^A > w$ will recruit foreign workers abroad, as in Figure 1.12. Countries with $w^A < w$ will export labor instead. The autarky wage depends on a number of factors. For example, an increase in a country's labor endowment, L^H, lowers its autarky wage rate and raises its net exports of labor. Countries that save more in the first period will tend to have higher autarky wages and higher net immigration.

The usual gains from trade are apparent in the figure: the GNP line lies above the autarky PPF except at **B**, where it is tangent. This result is based, however, on a representative-agent setup in which international labor-market integration has no distributional effects. When the real wage falls as a result of trade, for example, the representative agent gains more in his role as a capitalist than he loses in his role as a laborer.

Even in an economy of heterogenous individuals, everyone can gain from trade provided lump-sum side payments are made to redistribute its benefits. Such side payments are rarely made in reality, so some economic groupings are likely to lose from international labor movements. These losses explain the fierce opposition immigration usually arouses in practice. We will return to immigration again in Chapter 7.

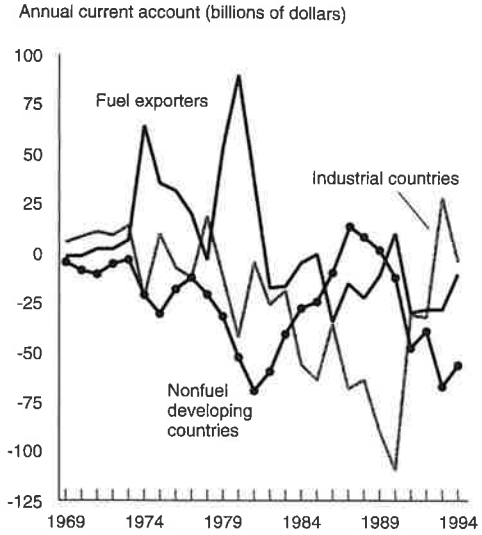


Figure 1.13
Global current account patterns, 1969–94

Application: Energy Prices, Global Saving, and Real Interest Rates

To conclude this chapter we revisit—not for the last time—the vexing question of explaining the wide swings in real interest rates of recent decades. An earlier application gave reasons to doubt the hypothesis that increases in actual or anticipated investment productivity were the primary factor behind the sharp rise in real interest rates in the early 1980s. A glance at Figure 1.9 shows that while real interest rates have been high since the 1980s in comparison with the 1960s, they were unusually low from 1974 to 1979. We can get clues about the key factors moving real interest rates by looking at their behavior, not just in the early 1980s, but over the entire period starting in the early 1970s.

A striking feature of Figure 1.9 is the sharpness with which rates decline in the early 1970s. The major shock to the world economy in that period was the decision by the world oil cartel, the Organization of Petroleum Exporting Countries (OPEC), to quadruple the dollar price of oil. Figure 1.13 shows the immediate impact of this event on the current accounts of three major country groups, the fuel exporters, other developing countries, and the industrial countries. The industrial countries ran a small and temporary deficit, the nonfuel developing countries ran a larger (relative to their GDPs) and more persistent deficit, and the fuel-exporting group moved to a massive surplus.

The major beneficiaries of the oil shock included countries like Kuwait and Saudi Arabia, which were unable to raise their spending quickly in line with the massive increase in their wealth. This inability explains the fuel exporters' current account surplus, and also suggests an explanation for the initial decline in real interest rates. OPEC's price hike, by worsening the rest of the world's terms of trade, caused a transfer of current income from its customers to itself. It caused a similar transfer of lifetime wealth. Since the OPEC countries as a group had a lower marginal propensity to spend out of wealth, their consumption rose by less than the fall in non-OPEC consumption. As a result, the world saving curve pictured in Figure 1.10 shifted to the right. To the extent that the oil price increase discouraged investment outside OPEC, the investment curve shifted to the left. Both shifts helped push the world interest rate down.[24]

The fuel exporters' external surplus shot up again in 1979 when a second OPEC shock followed the Iranian revolution. They were able to raise their spending more rapidly this time. As Figure 1.13 shows, fuel exporters as a group had a roughly balanced current account by 1981, the year real interest rates rose above their levels of the 1960s.

The conclusive disappearance of the OPEC surplus is not the only new trend that begins in 1981. Also, the industrial economies go into a protracted current account deficit (much of which is accounted for by the United States). The coincidence of these events with the rise in real interest rates is intriguing, and suggests that a look at the saving trends underlying the current account patterns could throw light on the real interest rate mystery.

Table 1.4 suggests that differences and shifts in countries' saving patterns might go a long way toward explaining the broad swings in world real interest rates. Saving rates in the industrial world declined substantially between the 1970s and 1980s. Furthermore, the saving rate of the fuel-exporting group, initially much higher than that of the rest of the world, dropped precipitously between the same two decades. Finally, other developing countries, which are smaller actors in the world capital market than the industrial group, register only a minor rise in saving in the early 1980s followed by a bigger increase later on. In terms of Figure 1.10, a leftward shift of the world saving schedule in the early 1980s would help explain the simultaneous reduction in global saving and high real interest rates of the decade. The exceptionally high saving rate of the fuel-exporting group during the 1970s helps explain why unusually low real interest rates followed the first oil shock. Other factors to be discussed later in this book, notably shifts in monetary policies, were at work, too, particularly in determining the year-to-year interest-

24. This effect on the intertemporal trade terms of income redistribution between economies with different intertemporal spending patterns is an example of the classical *transfer problem* of trade theory, which we revisit in detail in Chapter 4. Sachs (1981) reviews international adjustment to the oil shocks of the 1970s.

Table 1.4
Gross Saving as a Percent of GDP (period average)

Country Group	1973–80	1981–87	1988–94
Industrial	23.5	20.9	20.5
Fuel exporting	42.0	20.2	19.7
Nonfuel developing	22.4	22.9	26.2

Source: International Monetary Fund, *World Economic Outlook*, October 1995.

rate movements in Figure 1.9. One must always be cautious in interpreting simple correlations, and we do not mean the analysis here as anything more than suggestive. Nevertheless, long-term movements correspond surprisingly well to a story based on inter-country and intertemporal variation in the supply of savings.

What are the underlying causes of the secular decline in world saving shown in Table 1.4? That is a topic we will examine in considerable detail in Chapter 3. ■

Appendix 1A Stability and the Marshall-Lerner Condition

A market is stable in the Walrasian sense if a small increase in the price of the good traded there causes excess supply while a small decrease causes excess demand. This simple definition of stability is obviously inapplicable to a general-equilibrium, multimarket context: the meaning of stability and its significance are the subject of a large literature, see Arrow and Hahn (1971). In the two-country model of section 1.3, however, where only a single price—the interest rate—needs to be determined, the simple Walrasian stability condition is easily interpreted in a general-equilibrium setting. That condition, as we show here, is equivalent to an elasticity inequality with a long history in static international trade theory, the *Marshall-Lerner* condition.[25]

With saving and investment written as functions of the world interest rate, the global equilibrium condition is

$$S_1(r) + S_1^*(r) = I_1(r) + I_1^*(r).$$

The condition defining Walrasian stability in the market for world savings is that a small rise in r should lead to an excess supply of savings:

$$S_1'(r) + S_1^{*\prime}(r) - I_1'(r) - I_1^{*\prime}(r) > 0. \tag{39}$$

To relate this inequality to the Marshall-Lerner elasticity condition, let's assume for the moment that Home is an importer of first-period consumption and an exporter of second-period consumption. (Home has a current account deficit, followed by a surplus. An equilibrium with zero current accounts always is stable.) Home's imports are denoted by $IM_1 \equiv C_1 + I_1 - Y_1 = I_1 - S_1 > 0$, its exports by $EX_2 \equiv Y_2 - C_2 - I_2 > 0$. The condition of "balanced trade" in this model, that the value of imports equal that of exports, is

25. For a detailed discussion of stability conditions in international trade, see R. Jones (1961).

$$IM_1 = \left(\frac{1}{1+r}\right) EX_2,$$

where trade-flow values are in units of date 1 consumption. This condition always holds because it is nothing more than a rearrangement of Home's intertemporal budget constraint. For Foreign, $IM_2^* \equiv C_2^* + I_2^* - Y_2^* > 0$ and $EX_1^* \equiv Y_1^* - C_1^* - I_1^* = S_1^* - I_1^* > 0$.

Foreign's intertemporal budget constraint implies its current account is $S_1^* - I_1^* = IM_2^*/(1+r)$. We therefore can express stability condition (39) as

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[\left(\frac{1}{1+r}\right) IM_2^*(r) - IM_1(r)\right] > 0. \tag{40}$$

To express this inequality in terms of import elasticities, define

$$\zeta \equiv -\frac{(1+r)IM_1'(r)}{IM_1(r)}, \qquad \zeta^* = \frac{(1+r)IM_2^{*\prime}(r)}{IM_2^*(r)}$$

(The elasticity ζ is defined to be positive when a rise in r reduces Home's date 1 current account deficit.) Since the initial interest rate is an equilibrium rate such that $IM_1 = IM_2^*/(1+r)$, eq. (40) can be rewritten as

$$\zeta + \zeta^* > 1. \tag{41}$$

The Walrasian stability condition holds if and only if the sum of the Home and Foreign price elasticities of import demand exceeds 1. International trade theorists call condition (41) the Marshall-Lerner condition. When Home happens to be the exporter of date 1 consumption, rather than the importer, (41) still characterizes the Walras-stable case, but with import elasticities defined so that Home's and Foreign's roles are interchanged.

Exercises

1. *Welfare and the terms of trade.* Let the representative individual in a small open economy maximize $U(C_1, C_2)$ subject to $C_1 + C_2/(1+r) = Y_1 + Y_2/(1+r)$, where Y_1 and Y_2 are fixed.

   (a) Show that the intertemporal Euler equation takes the form $\partial U(C_1, C_2)/\partial C_1 = (1+r)\partial U(C_1, C_2)/\partial C_2$.

   (b) Use the Euler condition together with the (differentiated) budget constraint to compute the total derivative

   $$\frac{\mathrm{d}U(C_1, C_2)}{\mathrm{d}r} = \frac{\partial U(C_1, C_2)}{\partial C_2}(Y_1 - C_1).$$

   (c) Explain why the answer in b implies that a country benefits from a rise in the world interest rate if and only if its terms of intertemporal trade improve.

   (d) Let $W_1 \equiv Y_1 + Y_2/(1+r)$ (that is, W_1 is lifetime wealth in units of date 1 consumption). Show that a small percentage gross interest rate increase of $\hat{r} = \mathrm{d}r/(1+r)$ has the same effect on lifetime utility as a lifetime wealth change of $\mathrm{d}W_1 = \hat{r}(Y_1 - C_1)$.

2. *Logarithmic case of the two-country endowment model.* Consider the pure endowment model, in which equilibrium holds when $S_1 + S_1^* = 0$. Home's utility function is

   $$U_1 = \log C_1 + \beta \log C_2. \tag{42}$$

   Foreign has an analogous log utility function, with its consumption levels and time-preference factor distinguished by asterisks. Governments consume no resources.

   (a) Home receives perishable endowments Y_1 and Y_2 in the two periods. Show that the Home date 1 consumption function is a function of r,

   $$C_1(r) = \frac{1}{1+\beta}\left(Y_1 + \frac{Y_2}{1+r}\right).$$

   [This equation shows a general property of logarithmic preferences that we will use many times in this book: expenditure shares on the available goods are constants that correspond to the relative weights on the logarithmic summands in U_1. Thus spending on date 1 consumption, C_1, is a fraction $1/(1+\beta)$ of lifetime income (measured in date 1 consumption units), and spending on date 2 consumption, $C_2/(1+r)$, is a fraction $\beta/(1+\beta)$ of lifetime income. This property follows from the unitary intertemporal substitution elasticity.]

   (b) Show that Home saving is

   $$S_1(r) = Y_1 - C_1(r) = \frac{\beta}{1+\beta}Y_1 - \frac{1}{(1+\beta)(1+r)}Y_2.$$

   (c) Compute the equilibrium world interest rate.

   (d) Check that it lies between the autarky rates r^{A} and r^{A*}.

   (e) Confirm that the country with an autarky interest rate below r will run a current account surplus on date 1 while the one with an autarky rate above r will run a deficit.

   (f) How does an increase in Foreign's rate of output growth affect Home's welfare? Observe that a rise in the ratio Y_2^*/Y_1^* raises the equilibrium world interest rate. Then show that the derivative of U_1 with respect to r is

   $$\frac{\mathrm{d}U_1}{\mathrm{d}r} = \frac{\beta}{1+r}\left[\frac{r - r^{A}}{(1+r) + \beta(1+r^{A})}\right].$$

   What is your conclusion?

3. *Adding investment to the last exercise.* Assume date 2 Home output is a strictly concave function of the capital stock in place multiplied by a productivity parameter,

   $$Y_2 = A_2 K_2^{\alpha}$$

   ($\alpha < 1$), with a parallel production function in Foreign. (Date 1 outputs are exogenous because they depend on inherited capital stocks.)

   (a) Investment is determined so that the marginal product of capital equals r. Show that this equality implies

$$K_2 = \left(\frac{\alpha A_2}{r}\right)^{\frac{1}{1-\alpha}}.$$

(b) Show that Home's **II** schedule can be written as

$$I_1(r) = K_2 - K_1 = \left(\frac{\alpha A_2}{r}\right)^{\frac{1}{1-\alpha}} - K_1.$$

(c) Derive Home's date 1 consumption function, and show it can be written as

$$C_1(r) = \frac{1}{1+\beta}\left(Y_1 - I_1 + \frac{Y_2 - I_2}{1+r}\right).$$

(d) Using $I_2 = -K_2$ and the results of parts a–c, explain why

$$C_1(r) = \frac{1}{1+\beta}\left[K_1 + Y_1 + \frac{(1-\alpha)}{1+r}\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}} A_2^{\frac{1}{1-\alpha}}\right],$$

and conclude that the equation for **SS** is the upward-sloping curve

$$S_1(r) = Y_1 - \frac{1}{1+\beta}\left[K_1 + Y_1 + \frac{(1-\alpha)}{1+r}\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}} A_2^{\frac{1}{1-\alpha}}\right].$$

4. *Problem on $\sigma = 0$.* The individual has an isoelastic period utility function and exogenous endowments. This exercise considers the limit as $\sigma \to 0$.

(a) Show that the Euler equation (25) approaches $C_2 = C_1$, so that a flat consumption path is chosen irrespective of the market interest rate r.

(b) Derive the consumption function for this case,

$$C_1 = \left(\frac{1+r}{2+r}\right) Y_1 + \left(\frac{1}{2+r}\right) Y_2.$$

Show that $C_2 = C_1$ using this consumption function and the current account identity.

(c) Calculate that

$$\frac{dC_1}{dr} = \frac{Y_1 - Y_2}{(2+r)^2}.$$

What is your interpretation?

(d) Does zero intertemporal substitutability necessarily imply a literally constant consumption path, as in parts a–c, under all possible preference assumptions? [Hint: Suppose lifetime utility is

$$U_1 = \frac{1}{1 - 1/\sigma}\left(C_1^{1-1/\sigma} + \beta^{1/\sigma} C_2^{1-1/\sigma}\right).$$

Show that as $\sigma \to 0$, we approach $C_2 = \beta C_1$, which corresponds to the consumption pattern chosen under the Leontief utility function $U_1 = \min\{\beta C_1, C_2\}$.]

5. *Endowment shifts and world interest rates.* In the two country endowment model of borrowing and lending, show algebraically that a rise in Y_1 or Y_1^* lowers r, whereas a rise in Y_2 or Y_2^* raises r.

6. *Future productivity shocks when current accounts are initially unbalanced.* Let Home have the production function $Y = AF(K)$, and Foreign the function $Y^* = A^*F^*(K^*)$, on each of two dates. Let the corresponding lifetime utility functions of residents be $U_1 = u(C_1) + \beta u(C_2)$ and $U_1^* = u(C_1^*) + \beta u(C_2^*)$, where $u(\cdot)$ is isoelastic. On date 1 the countries may borrow (lend) at the world interest rate r, determined by $S_1 + S_1^* = I_1 + I_1^*$. A Walras-stable world market with a single equilibrium is assumed.

(a) Suppose date 2 Home investment productivity, A_2, rises slightly. If Home has a date 1 surplus on current account and Foreign a date 1 deficit before the productivity rise, how does the change affect date 1 current accounts?

(b) Do the same exercise for a rise in A_2^* (Foreign's date 2 productivity), still assuming Home has a positive current account on date 1. Is the effect on Home's current account simply a mirror image of the answer to part a?

7. *Interest rates and saving with exponential period utility.* A country's representative individual has the exponential period utility function

$$u(C) = -\gamma \exp(-C/\gamma)$$

($\gamma > 0$) and maximizes $U_1 = u(C_1) + \beta u(C_2)$ subject to

$$C_1 + RC_2 = Y_1 + RY_2 = W_1$$

[where $R \equiv 1/(1+r)$].

(a) Solve for C_2 as a function of C_1, R, and β using the consumer's intertemporal Euler equation.

(b) What is the optimal level of C_1, given W_1, R, and β?

(c) By differentiating this consumption function C_1 (including differentiation of W_1) with respect to R, show that

$$\frac{dC_1}{dR} = -\frac{C_1}{1+R} + \frac{Y_2}{1+R} + \frac{\gamma}{1+R}\left[1 - \log(\beta/R)\right].$$

(d) Calculate the inverse elasticity of marginal utility, $-u'(C)/Cu''(C)$, for the exponential utility function. [It is a function $\sigma(C)$ of consumption, rather than being a constant.]

(e) Show that the derivative dC_1/dR calculated in part c above can be expressed as

$$\frac{dC_1}{dR} = \frac{\sigma(C_2)C_2}{1+R} - \frac{C_2}{1+R} + \frac{Y_2}{1+R}.$$

Interpret the three additive terms that make up this derivative.

8. *The optimal borrowing tax.* Return to section 1.5's model of the optimal tax on foreign borrowing, but assume a general utility function $U(C_1, C_2)$.

(a) Show that the optimal ad valorem borrowing tax [such that $(1+\tau)(1+r^{\tau})$ is the gross interest rate domestic residents face when the world rate is $1+r^{\tau}$] is given by $\tau = 1/(\zeta^* - 1)$, where ζ^* is the elasticity with respect to $1+r$ of Foreign's demand for imports of date 2 consumption. (The elasticity ζ^* is defined formally in the chapter appendix.)

(b) Explain why $\zeta^* - 1 > 0$ at the world interest rate associated with the optimal tax.

2 Dynamics of Small Open Economies

The last chapter's two-period model illuminates the basic economics of international borrowing and lending. Despite its important lessons, it misses many important issues that cannot easily be condensed within a two-period horizon. What are the limits on a growing country's foreign debt? What if capital-stock changes must take place over several time periods, or if consumption includes durable goods? How do expected movements in future short-term interest rates affect the current account? Perhaps most importantly, the theories we have developed so far cannot meaningfully be adapted to real-world data without extending them to a multiperiod environment.

In most of the chapter we will simplify by assuming not only that there are many time periods, but that there is no definite end to time. This abstraction may seem extreme: after all, cosmologists have developed theories predicting that the universe as we know it will end in 60 to 70 billion years! That grim prophecy notwithstanding, there are two defenses for an infinite-horizon assumption. First, there is genuine uncertainty about the terminal date. The relevant cosmological theories yield only approximations, and theories, after all, are revised in light of ongoing scientific discovery. Second, other things being the same, the behavior of an infinite-horizon economy will differ only trivially from that of a finite-horizon one when the terminal date is very distant.

It is quite another matter to assume, as we do in this chapter, that *individuals* live forever. Once again, two defenses are offered. First, uncertainty about length of life can act much like an infinite horizon in blurring one's terminal planning date. Second, when people care about their offspring, who, in turn, care about theirs, an economy of finite-lived individuals may behave just like one peopled by immortals. Chapter 3 explores this possibility rigorously.

In this chapter we focus on the case of a small open economy inhabited by a representative individual with an infinite horizon. The economy is small in the sense that it takes the path of world interest rates as exogenous. Abstracting from global general-equilibrium considerations makes sense here, since they are not important for many questions and would unnecessarily complicate the analysis. Besides, for the vast majority of countries in the world, the small-country assumption is the right one. There are really only a few economies large enough that their unilateral actions have a first-order impact on world interest rates. However, in studying the small-country case, one does have to be on guard against scenarios that strain the paradigm. For example, if a small country's Gross Domestic Product were perpetually to grow faster than world GDP, the country would eventually become large! Similarly, if the country is a high saver with ever-growing net foreign assets, the assumption that it always faces a fixed world interest rate may again become strained. We will alert the reader on occasions where such issues arise, though we defer full general equilibrium analysis of the particular class of models considered here until we take up global growth in Chapter 7.